STUDENT SKILL AND PLACEMENT DEVELOPMENT TRACKING

A PROJECT REPORT

Submitted by

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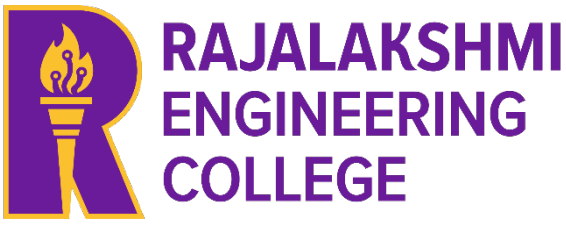
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "STUDENT SKILL AND PLACEMENT DEVELOPMENT TRACKING" is the bonafide work of REVATHY K (2414101080), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, REVATHY K (2414101080), hereby declare that the project report titled “STUDENT SKILL AND PLACEMENT DEVELOPMENT TRACKING”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03 November 2026

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ACKNOWLEDGEMENT

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I owe my sincere thanks to my course faculty, M. L. Alphin Ezhil Manuel, Assistant Professor (SG), for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

REVATHY K

ABSTRACT

Campus placement preparation in most engineering colleges is still managed through scattered spreadsheets, paper forms and messaging groups. Student profiles, skill scores, company details, drive schedules and interview outcomes are kept in separate files, so the placement cell spends considerable effort consolidating data, checking eligibility and preparing reports, while students have little visibility of their own progress.

This project presents Student Skill and Placement Development Tracking, a low-code application built on Zoho Creator for the Department of Computer Science and Business Systems at Rajalakshmi Engineering College. Zoho Creator was chosen because its form builder, built-in hosting, workflow engine and role-based security allow a complete business application to be developed and changed quickly without a conventional development stack. The application contains six forms (Student Registration, Skill Assessment, Company Registration, Job Drive, Job Application, and Interview and Placement), seven reports and a Student Dashboard page with key figures and charts. Form workflows synchronise every registered company to Zoho CRM, check each job application against the minimum CGPA of the drive and e-mail a placement confirmation to the student, a schedule workflow sends interview reminders, and four roles (Admin, Placement Officer, Faculty and Student) with matching permission profiles control access to data.

The application replaces repeated manual data entry with a single connected data set, so that eligibility checks, shortlisting, interview tracking and placement reports are produced from live records rather than separate spreadsheets. The dashboard gives the placement cell an immediate view of registered students, participating companies, drives, applications and placed students. The report also describes the testing carried out role by role, the deployment and security configuration, and the limits of the low-code approach observed during development.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Placement Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	STUDENT SKILL AND PLACEMENT DEVELOPMENT TRACKING
Application Domain	Education – Training and Placement
Student Name & Register No.	REVATHY K – 2414101080
Zoho Creator Edition	Trial edition (the application header shows “Trial expires in 14 days” on 01 November 2026)
Application Name (Link Name)	Student Skill And Placement Tracking System | link name: student-skill-and-placement-tracking-system
Published / Live URL	https://revathyk2024csbsrajal454.zohocreatorportal.in
Demo Credentials (per role)	Administrator, Placement Officer, Faculty and Student logins are provided to the examiner at the demonstration (passwords are not printed in this report)
Number of Forms / Subforms	6 / 0
Number of Reports (by type)	7 – List: 6 · Kanban: 0 · Calendar: 0 · Pivot/Chart: 0 (charts are on the dashboard page) · Others: 1 (Student Skill Assessment Report, card layout)
Number of Pages / Dashboards	1 – Student Dashboard
Number of Workflows	Form: 3 (Sync Company to Zoho CRM, Job Eligibility Check, Placement Status Notification) | Schedule: 1 (Interview Reminder) | Approval: 1 (see Section 5.6)
Deluge Functions / Custom Actions	Deluge scripts: 3 (DS-01 to DS-03) | Custom functions: none (no standalone function)
Integrations	Zoho CRM (company synchronisation)
Roles & Profiles Defined	Admin, Placement Officer, Faculty, Student (profiles: Admin Access, Placement Officer Access, Faculty Access, Student Access)
Environments Used	Development → Stage → Production (version 1.0 published on 01 November 2026; see Section 7.1)
Date of Production Deployment	01 November 2026 (version 1.0 in Stage and Production)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
DS	Deluge Script
ER	Entity–Relationship
FR	Functional Requirement
HoD	Head of the Department
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
OAuth	Open Authorization
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Campus placement is one of the most important outcomes of an engineering programme. Students must build technical and soft skills, keep their academic record in good standing, apply to companies whose eligibility criteria they satisfy, and then go through aptitude tests, technical interviews and HR rounds. The training and placement cell, the faculty and the students are all involved, and each group needs a different view of the same information.

This project considers the placement process of the Department of Computer Science and Business Systems at Rajalakshmi Engineering College, Chennai. Student details, certifications and skill scores are collected in spreadsheets, company contacts are kept in emails and separate files, and students are told about drives and shortlists through messaging groups. Four groups of people take part: the Placement Officer, who coordinates companies and drives; the Faculty, who monitor skill development; the Students, who register and apply; and the Admin, who manages the system.

1.2 Problem Statement

There is no single system in which a student's skills, a company's requirements, the applications made and the final placement outcome are connected. As a result, the placement cell repeats data entry, eligibility is checked by hand, and management cannot see the placement position without preparing a report manually. The specific pain points are listed below.

- Student, company and drive data are spread over separate spreadsheets and emails, which leads to duplication and inconsistent records.
- Eligibility checking and shortlisting depend on manual comparison of CGPA, skills and department.
- Faculty and students have no real-time view of skill development, application status or interview results.
- Placement reports for the HoD and management take time to prepare and become outdated quickly.

- Anyone holding a shared spreadsheet can see or change every record, because there is no role-based access control.

1.3 Objectives

- To design a normalised data model and forms for students, skill assessments, companies, job drives, job applications and interview and placement records.
- To automate company synchronisation, eligibility checking of applications, placement notifications and interview reminders using workflows and Deluge scripting.
- To provide role-based dashboards and reports for the Admin, Placement Officer, Faculty and Student roles.
- To integrate the application with Zoho CRM for keeping registered company records in a customer relationship system.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In scope: six forms (Student Registration, Skill Assessment, Company Registration, Job Drive, Job Application, and Interview and Placement); seven reports; a Student Dashboard page with KPI panels and charts; form workflows that synchronise companies to Zoho CRM, check job eligibility by CGPA and send placement confirmation e-mails; a schedule workflow for interview reminders; and four user roles (Admin, Placement Officer, Faculty, Student) with permission profiles. The application is used through the Zoho Creator web interface, and the same application can be opened in the Zoho Creator mobile app.

Out of scope: online aptitude tests, automatic resume parsing, payment handling, and integration with the college ERP. These are listed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

Low-code development suits this problem for the following reasons [1], [2]. First, the data model is form-centred, which maps directly to the visual form builder of Zoho Creator, so six connected forms could be created without writing database code. Second, workflows, approvals and scheduled jobs are configured visually and extended

with Deluge only where needed, which shortens development time. Third, hosting, backup, mobile access and role-based security are provided by the platform, so the department does not have to maintain servers. Finally, because the application will be maintained by department staff rather than professional developers, a platform that a citizen developer can modify is appropriate.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing process, students give their details on a paper form or a shared spreadsheet. The placement cell merges these files, checks eligibility manually against the criteria sent by each company, and informs students about drives and shortlists through email and messaging groups. Interview results are noted in another spreadsheet, and the placement report is prepared by hand for the HoD and management. Figure 2.1 shows this flow.

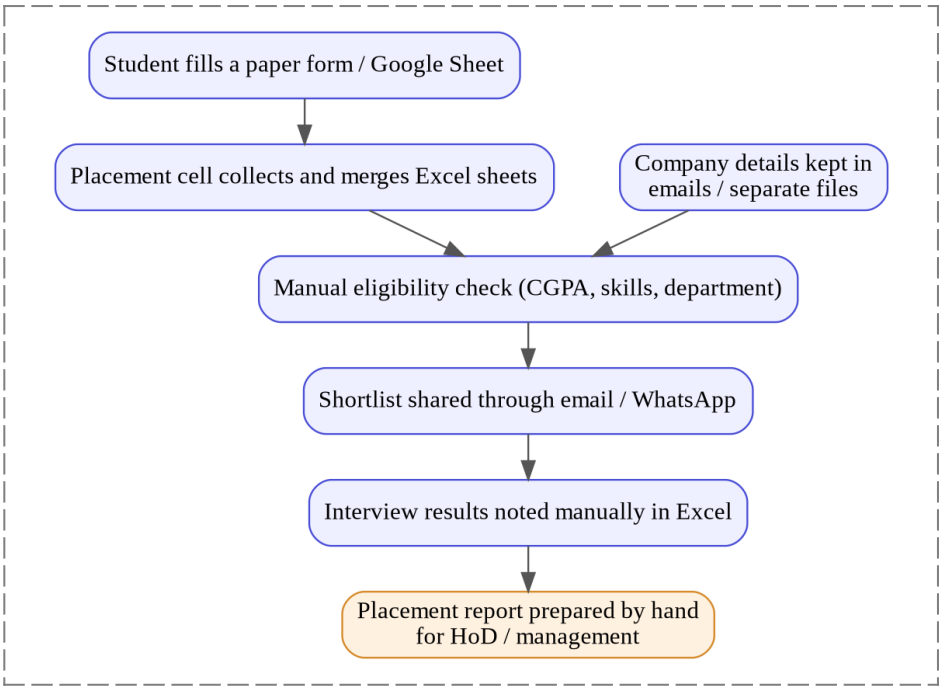


Figure 2.1: As-Is Process Flow of the Existing Placement Tracking Process

2.2 Limitations of the Existing System

- Data is duplicated across spreadsheets, so the same student or company may have conflicting entries.
- Eligibility checks and shortlisting are manual and error-prone.
- There is no approval trail or status tracking for applications.
- Reports are prepared by hand and are not available in real time.
- There is no access control: every user of a shared file can edit every record.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks (estimate) for design, coding, database set-up, hosting and testing	Approximately 3–5 days of part-time work (estimate)
Skills required	Front-end, back-end language, database, authentication and hosting skills	Form design, basic Deluge scripting and understanding of the business process
Hosting & infrastructure	Server, database, backups and SSL must be arranged and maintained	Hosted, backed up and secured by Zoho Creator
Mobile application	Separate mobile or responsive development	Available through the Zoho Creator mobile app without extra development
Security & access control	Login, roles and permissions coded and tested separately	Built-in users, roles, permission profiles and record-level criteria
Maintenance & change	Code change, testing and redeployment for every new field or report	Fields, reports and workflows are changed in the builder
Customisation limits	Practically unlimited	Limited by platform features, Deluge limits and page-builder options
Cost	Developer effort plus server and licence cost	Platform subscription with low development effort

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski, BPB Publications [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, benefits and limits, and how business users and developers share the work	Basis for the justification in Section 1.5 and the limitations in Section 8.4
2	Zoho Corporation, online documentation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, roles and integrations of Zoho Creator	Reference for every component built in Chapter 5

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	Zoho Corporation, online documentation [3]	Deluge Script Reference Guide	Describes the syntax, tasks and integration statements of Deluge	Reference for the scripting in Section 5.5
4	Zoho Corporation, learning resources [4]	Zoho Creator Learning Centre	Guided tutorials and examples of applications built in Zoho Creator	Guided the form, report and workflow design
5	Gartner, Inc. [5]	Magic Quadrant for Enterprise Low-Code Application Platforms	Evaluates enterprise low-code platform vendors and tracks the growth of the category	Supports the trends discussed in Section 2.5
6	Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, 2021 [6]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analyses practitioner discussions about low-code development and reports its characteristics and challenges	Informs the limitations in Section 8.4
7	M. A. Al Alamin et al., 2021 [7]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges	Analyses developer discussions about low-code and the challenges they raise	Informs the limitations in Sections 8.4 and 8.5

2.5 Emerging Trends in Low-Code Development

Four trends in low-code development influenced this application [5]. AI-assisted building, such as Zoho's Zia and the AI application builder, speeds up the creation of forms and scripts; the design was kept simple enough for such assistance to be added later (Section 9.2). Citizen development allows department staff, not only professional programmers, to maintain applications, which is why the application relies on configuration and short Deluge scripts. Hyper-automation, the combination of workflows, approvals and schedules, is reflected in the company synchronisation, approval process and scheduled follow-ups. Composable applications, where an application is joined to other services, are reflected in the integration with Zoho CRM.

2.6 Proposed System

The proposed system is a Zoho Creator application named Student Skill And Placement Tracking System. It keeps student profiles, skill assessments, companies, job drives, job applications and interview and placement results in connected forms, so that every record is entered once and reused (limitation 1). Reports and the Student Dashboard give live information instead of hand-made reports (limitation 4). A job eligibility script compares the student's CGPA with the minimum CGPA of the drive and records the result as the application status (limitation 3), workflows reduce hand-offs between people (limitation 2), and roles with permission profiles restrict each user to the data they need (limitation 5). Companies are also synchronised to Zoho CRM so that the contact details are available in a customer relationship system.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Admin	Administrator of the application with full access	Manage users, roles and settings; view and correct all data
Placement Officer	Coordinates companies, drives and placements	Register companies, create job drives, review application status, record interviews and placements, view the dashboard
Faculty	Monitors the skill development of students	Record and review skill assessments, view student reports
Student	Registers and applies to drives	Register profile, take part in skill assessments, submit job applications, track status

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a student to register with mandatory fields (name, student ID, department, year, college mail, CGPA and skill)	Student, Admin	High
FR-02	The system shall record the skill assessment scores of a registered student and show them in a report	Faculty, Admin	High
FR-03	The system shall allow a company to be registered with industry, location and HR contact details	Placement Officer, Admin	High
FR-04	The system shall synchronise every registered company to Zoho CRM on successful submission	System	Medium
FR-05	The system shall allow job drives to be created for registered companies	Placement Officer	High
FR-06	The system shall allow a student to submit a job application for a drive	Student	High
FR-07	The system shall compare the student’s CGPA with the minimum CGPA of the job drive and set the application status to Shortlisted or Rejected	System	High
FR-08	The system shall record the interview round, interview result, placement status and package offered	Placement Officer	High

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall e-mail a placement confirmation to the student when the placement status is Placed	System	Medium
FR-10	The system shall display a dashboard of registered students, companies, drives, applications and placements with charts	Admin, Placement Officer, Faculty	Medium
FR-11	The system shall send an interview reminder through a schedule workflow based on the Interview Date	System	Medium
FR-12	The system shall show each user only the forms, reports and pages permitted for their role	All roles	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 500 records shall load within 5 seconds
NFR-02	Security	Users shall access only the data permitted by their role
NFR-03	Usability	The application shall be usable on the web and in the Zoho Creator mobile app
NFR-04	Availability	The application shall run on the Zoho Creator cloud service, so no local server is required
NFR-05	Maintainability	Forms, reports and workflows shall be changeable by department staff in the Zoho Creator builder
NFR-06	Auditability	All record changes shall be traceable through the audit trail

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Trial edition (Professional features during the trial period)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman (optional, for testing API calls)
Other Zoho services	Zoho CRM
Mobile	Zoho Creator mobile app (Android / iOS), optional

Category	Specification
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

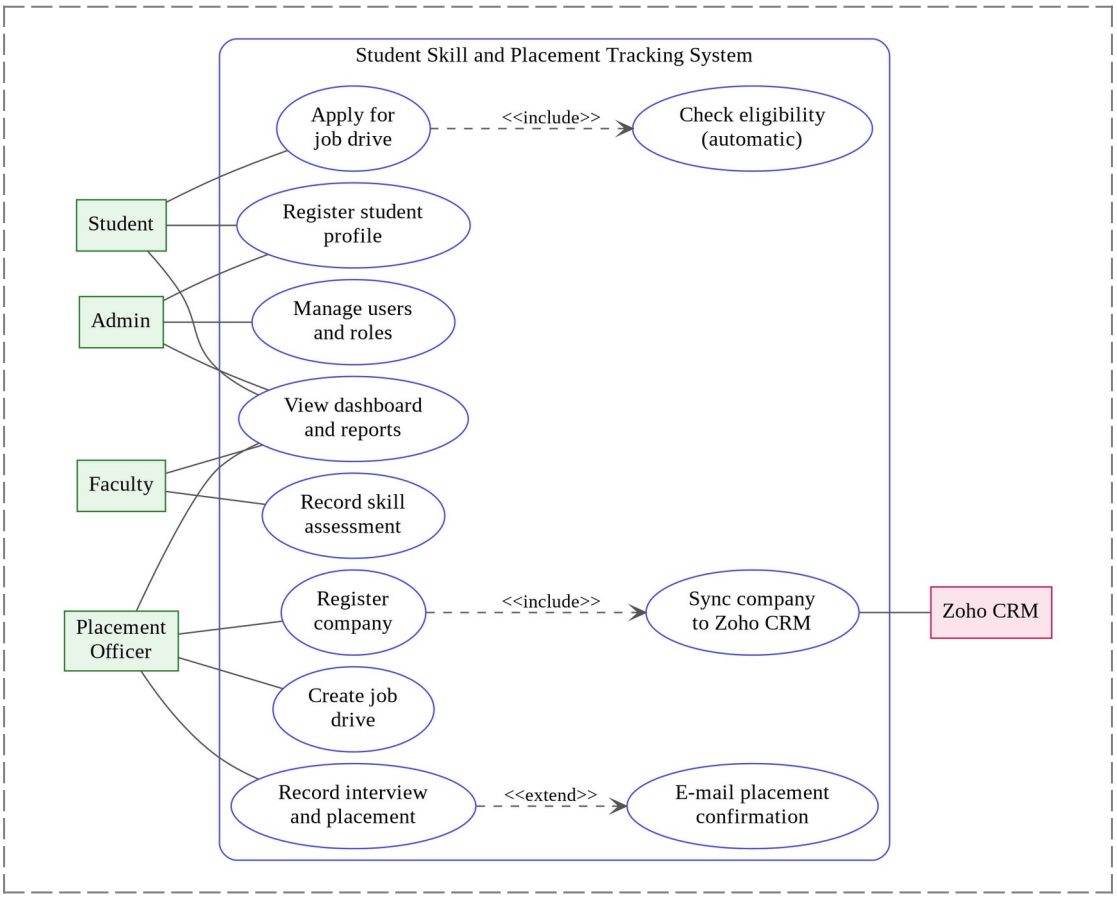


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Apply for Job Drive

Use Case ID	UC-01
Use Case Name	Apply for Job Drive
Actor(s)	Student; System (Job Eligibility Check workflow)
Pre-condition	The student is registered and the Placement Officer has created the job drive with its Minimum CGPA
Main Flow	1. The student opens Job Applications → Job Application. 2. The student selects the student record and the job drive. 3. The student submits the form. 4. The workflow Job Eligibility Check compares the student's CGPA with the Minimum CGPA of the drive. 5. The system sets Application_Status to Shortlisted.

Alternate Flow	If the student's CGPA is below the Minimum CGPA, the status is set to Rejected. If a mandatory field is empty, the form shows an error and nothing is saved.
Post-condition	The application is stored with the status Shortlisted or Rejected and appears in the Job Applications report.
Related FR	FR-06, FR-07

Table 3.6: Use Case Description – Record Interview and Placement

Use Case ID	UC-02
Use Case Name	Record Interview and Placement
Actor(s)	Placement Officer; System (notification workflow)
Pre-condition	A job application exists for the student
Main Flow	1. The Placement Officer opens Interview & Placement → Interview & Placement. 2. The officer selects the job application, student and company. 3. The officer enters job role, interview date, round, result, placement status and package offered. 4. The officer submits the form. 5. If Placement Status is Placed, the workflow Placement Status Notification e-mails the student.
Alternate Flow	If interview round, result or placement status is missing, the form shows an error.
Post-condition	The result is stored, the student is notified when Placed, and the Placed Students count on the dashboard is updated.
Related FR	FR-08, FR-09, FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

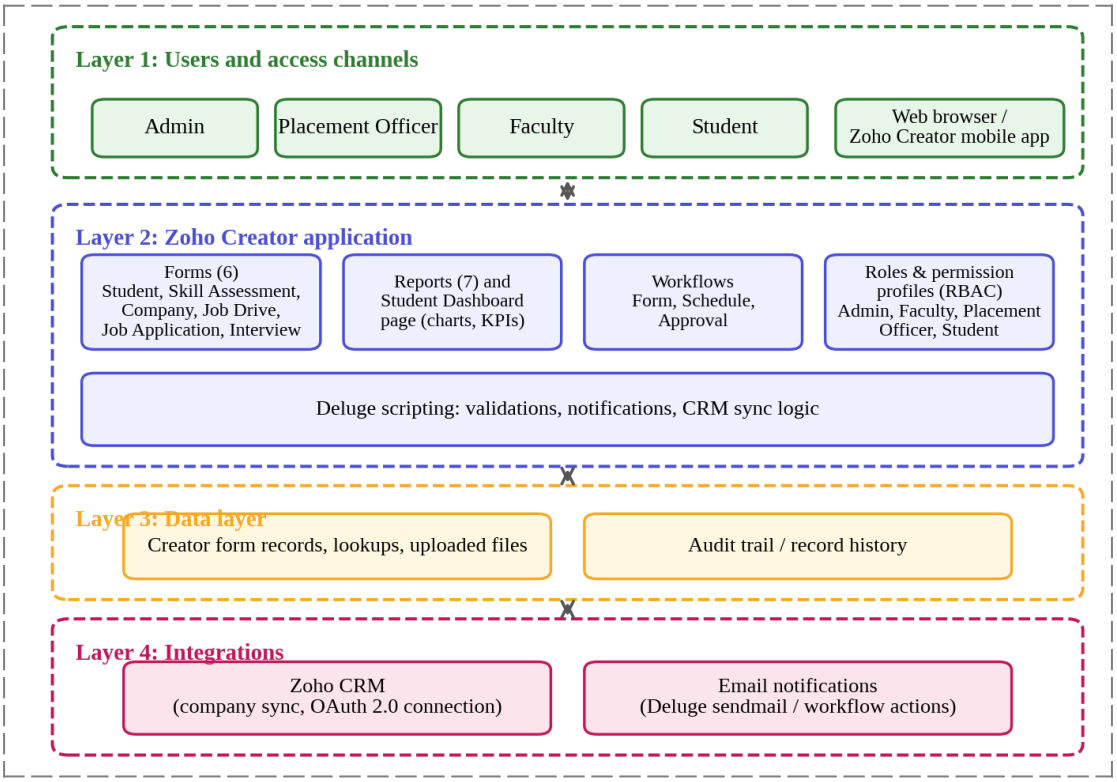


Figure 4.1: System Architecture of the Application

The architecture has four layers. In the first layer, the four roles reach the application through a web browser or the Zoho Creator mobile app. In the second layer, the Zoho Creator application holds six forms, seven reports, the Student Dashboard page, workflows (form, schedule and approval) and Deluge scripts, and the roles and permission profiles decide what each user can see and do. In the third layer, the Zoho Creator data layer stores form records, lookups and uploaded files such as resumes, and the audit trail records changes. In the fourth layer, the integration with Zoho CRM synchronises company records through an OAuth connection, and email notifications are sent through workflow actions.

4.2 Data Model

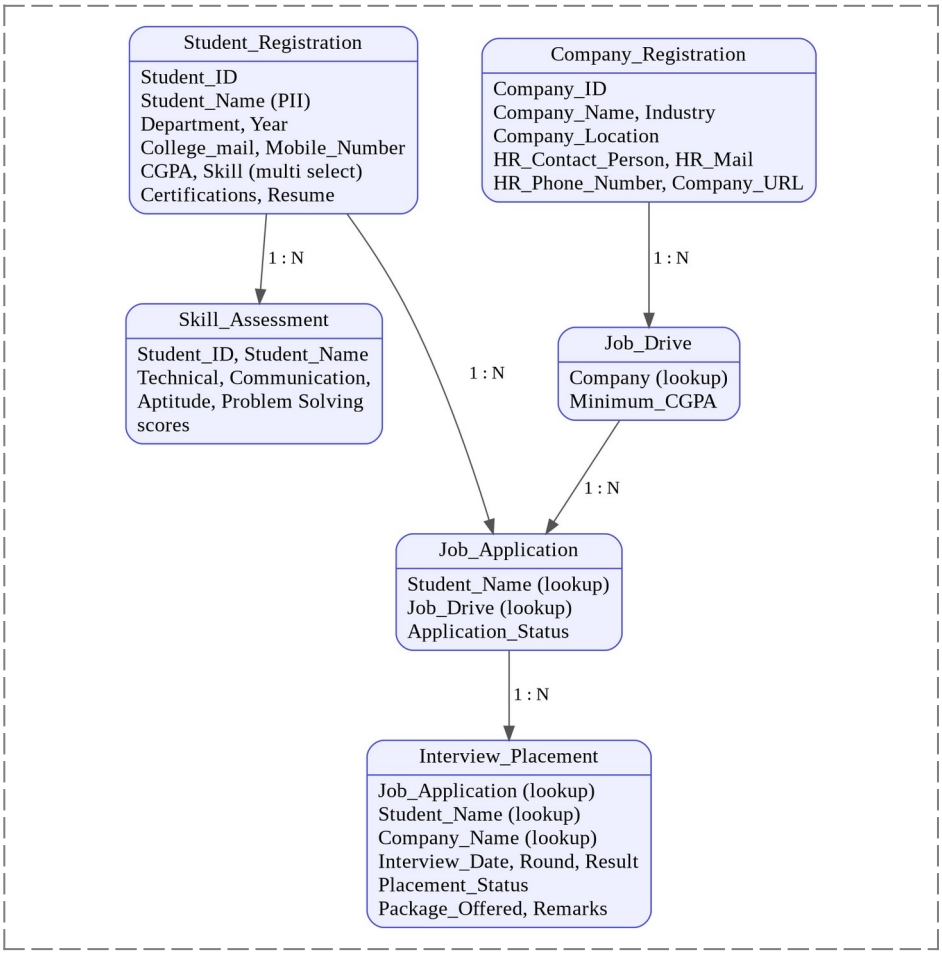


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Registration	Student_ID, Student_Name, Department, Year, College_mail, CGPA, Skill	Skill Assessment; Job Application; Interview & Placement	One-to-Many	Lookup (Student_Name) in Job Application and Interview & Placement
Skill Assessment	Student_ID, Student_Name, technical, communication, aptitude and problem-solving scores	Student Registration	Many-to-One	Linked by student ID and name

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Company Registration	Company_ID, Company_Name, Industry, Company_Location, HR_Mail	Job Drive; Zoho CRM	One-to-Many	Lookup in Job Drive; synchronised by form workflow
Job Drive	Company, Minimum_CGPA, Job_Role	Job Application	One-to-Many	Lookup (Job_Drive) in Job Application
Job Application	Student_Name (lookup), Job_Drive (lookup), Application_Status	Interview & Placement	One-to-Many	Lookup in Interview & Placement
Interview & Placement	Job Application, Student_Name, Company_Name, Interview_Round, Interview_Result, Placement_Status, Package_Offered	Job Application	Many-to-One	Lookup to Job Application

The remaining links are implemented as lookups: Job Drive → Company Registration (Company lookup), Skill Assessment → Student Registration (linked by Student ID and Student Name) and Interview & Placement → Job Application (Job_Application lookup).

4.3 Form Design

Table 4.2: Field Specification – Student Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Name	Yes	First Name, Last Name	Marked as personal data (PII)
Student ID	Student_ID	Single Line	Yes	—	Register number of the student
Department	Department	Drop Down	Yes	—	List of departments
Year	Year	Drop Down	Yes	—	Year of study
College mail	College_mail	Email	Yes	Email format	Official college mail ID

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Mobile Number	Mobile_Number	Phone	No	Country code +91	—
CGPA	CGPA	Decimal	Yes	Numeric	Used for eligibility
Skill	Skill	Multi Select	Yes	—	Skills selected from a list
Certifications	Certifications	Multi Line	No	—	Course and certificate details
Resume	Resume	File Upload	No	—	Resume document

Table 4.3: Field Specification – Skill Assessment Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Single Line	Yes	—	Identifies the student
Student Name	Student_Name	Name	Yes	First Name, Last Name	—
Technical skill score	Technical_skill_score	Number	Yes	Numeric	Technical skill score
Communication score	Communication_score	Number	Yes	Numeric	Communication score
Aptitude score	Aptitude_score	Number	Yes	Numeric	Aptitude score
Problem Solving score	Problem_Solving_score	Number	Yes	Numeric	Problem-solving score
Overall Score	Overall_Score	Decimal (calculated by workflow)	—	—	Shown as 76.00 in the report

Table 4.4: Field Specification – Company Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Company ID	Company_ID	Auto Number	—	System-generated	Shown in All Company Registrations
Company Name	Company_Name	Single Line	Yes	—	Name of the company

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Industry	Industry	Drop Down	No	—	IT/Software, Large Enterprise, Other
Company Location	Company_Location	Single Line	No	—	—
HR Contact Person	HR_Contact_Person	Single Line	No	—	—
HR Mail	HR_Mail	Email	Yes	Email format	Sent to Zoho CRM in the description
HR Phone Number	HR_Phone_Number	Phone	No	Country code +91	Sent to Zoho CRM as Phone
Company URL	Company_URL	URL	No	URL format	Company website

Table 4.5: Field Specification – Job Drive Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Company	Company	Lookup	Yes	—	Company of the drive
Minimum CGPA	Minimum_CGPA	Decimal	Yes	Numeric	Used by DS-02
Job Role	Job_Role	Single Line	Yes	—	Shown as the name of the drive in lookups (for example java developer)

Table 4.6: Field Specification – Job Application Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Job Drive	Job_Drive	Lookup	Yes	—	Lookup to Job Drive
Student Name	Student_Name	Lookup	Yes	—	Lookup to Student Registration
Application Status	Application_Status	Drop Down	No	Set by DS-02	Shortlisted or Rejected
Application Date	Application_Date	Date	No	dd-MMM-yyyy	Shown as 01-Nov-2026 in the report

Table 4.7: Field Specification – Interview & Placement Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Job Application	Job_Application	Lookup	Yes	Selects the application	Links to Job Application form
Student Name	Student_Name	Lookup	No	—	Student attending the interview
Company Name	Company_Name	Lookup	No	—	Company conducting the interview
Job Role	Job_Role	Single Line	No	—	—
Interview Date	Interview_Date	Date	No	dd-MMM-yyyy	Trigger of Interview Reminder
Interview Round	Interview_Round	Drop Down	Yes	—	Round of the interview
Interview Result	Interview_Result	Drop Down	Yes	—	Result of the round
Placement Status	Placement_Status	Drop Down	Yes	—	Placed triggers DS-03
Package Offered	Package_Offered	Decimal	No	Numeric	Offered package
Remarks	Remarks	Multi Line	No	—	—

4.4 Process and Workflow Design

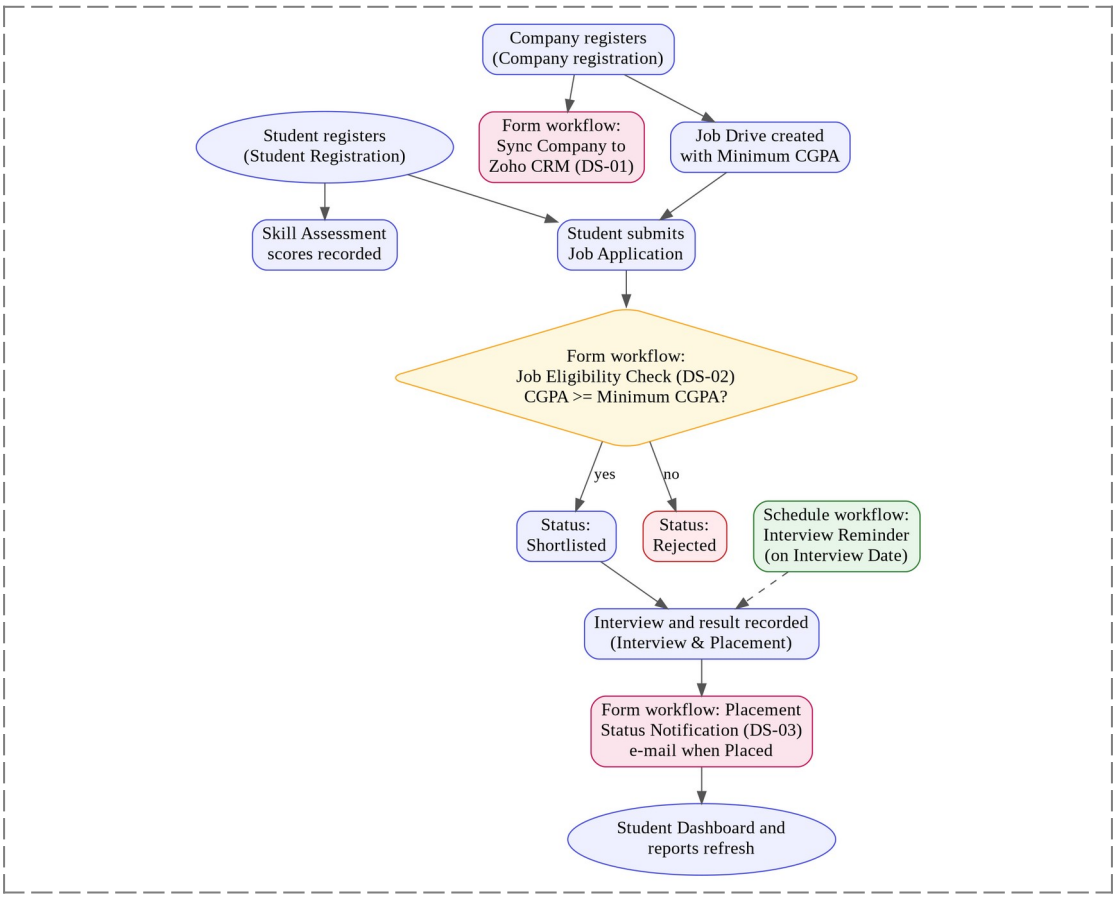


Figure 4.3: Workflow / Process Flow for the Placement Process

4.5 User Interface and Navigation Design

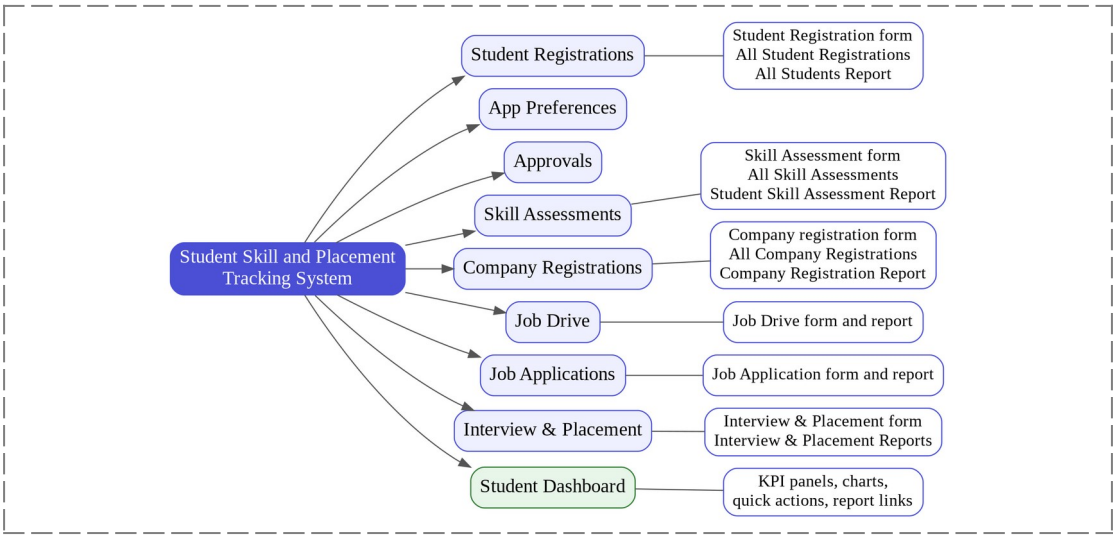


Figure 4.4: Navigation Structure of the Application

The application opens on the Student Dashboard, which is also its home page. The wireframe of the dashboard is shown with the finished page in Figure 5.1.

4.6 Access Control Design

Table 4.8: Role–Permission Matrix

Component	Admin	Placement Officer	Faculty	Student
Form: Student Registration	A E V D	V	V	A E V (own)
Form: Skill Assessment	A E V D	V	A E V	V (own)
Form: Company Registration	A E V D	A E V	V	—
Form: Job Drive	A E V D	A E V D	V	V
Form: Job Application	A E V D	E V	V	A V (own)
Form: Interview & Placement	A E V D	A E V	V	V (own)
Report: All reports	V	V	V	V (own)
Page: Student Dashboard	V	V	V	V

The matrix above is the design target for the permission profiles described in Section 5.8.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, Student Skill And Placement Tracking System, was created in Zoho Creator from scratch. The left-hand menu is organised into sections: Student Registrations, App Preferences, Approvals, Skill Assessments, Company Registrations, Job Drive, Job Applications, Interview & Placement and Student Dashboard. Each form section contains the form and its reports, and the Student Dashboard is the home page that opens first (Figure 5.1). A dark navy menu with an indigo accent colour is used as the theme.

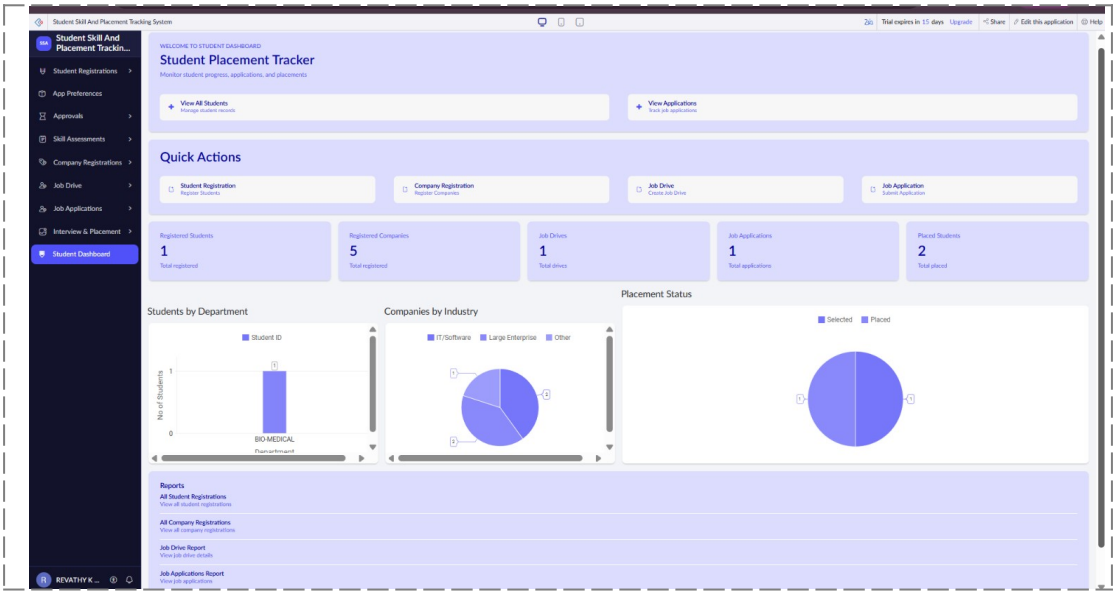


Figure 5.1: Application Home (Student Dashboard) with Navigation Sections

5.2 Forms and Data Capture

Six forms were created in the Form Builder. The fields follow the information that the placement cell collects today, and mandatory settings prevent incomplete records from being submitted.

Student Registration captures the student name (Name field with first and last name, marked as personal data), Student ID, Department and Year (drop downs), college mail (Email), mobile number (Phone with +91 country code), CGPA (Decimal), skill (Multi Select), certifications (Multi Line) and resume (File Upload). Name, Student ID, Department, Year, college mail, CGPA and Skill are mandatory (Figure 5.2, Figure 5.3).

Skill Assessment records the technical, communication, aptitude and problem-solving scores of a student together with the student ID and name; the scores appear in the Student Skill Assessment Report (Figure 5.7). The form is shown in design mode in Figure 5.4, with the labels placed above the fields.

Company registration records the company name and HR mail (mandatory), industry, location, HR contact person, HR phone number and company URL (Figure 5.5). A successful submission starts the workflow that synchronises the company to Zoho CRM.

Job Drive creates a placement drive for a registered company and stores the Minimum CGPA that a student needs to be shortlisted. The drive is identified by its Job Role (for example java developer), which is the value shown when a student selects the drive.

Job Application lets a student apply to a drive. It has lookups to Student Registration and Job Drive and an Application_Status field, which the Job Eligibility Check script (DS-02) fills as Shortlisted or Rejected. Application_ID is generated automatically and Application_Date records the date of the application (for example 01-Nov-2026), as shown in the All Job Applications report.

Interview & Placement links to a job application and records the student, company, job role, interview date, interview round, interview result, placement status, package offered and remarks. Job Application, Interview Round, Interview Result and Placement Status are mandatory (Figure 5.6).

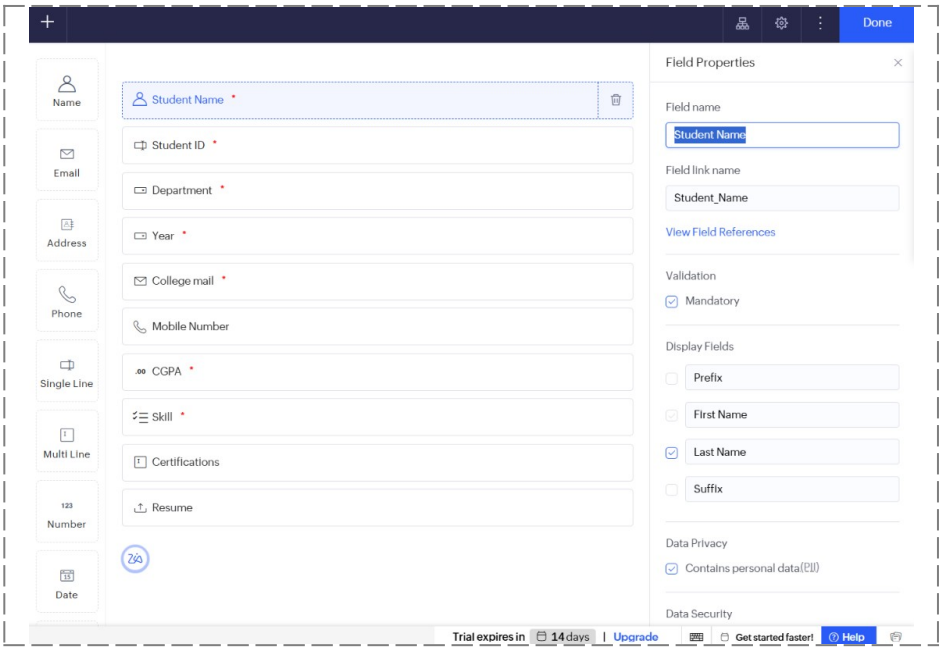


Figure 5.2: Student Registration Form in the Form Builder (Field Properties of Student Name)

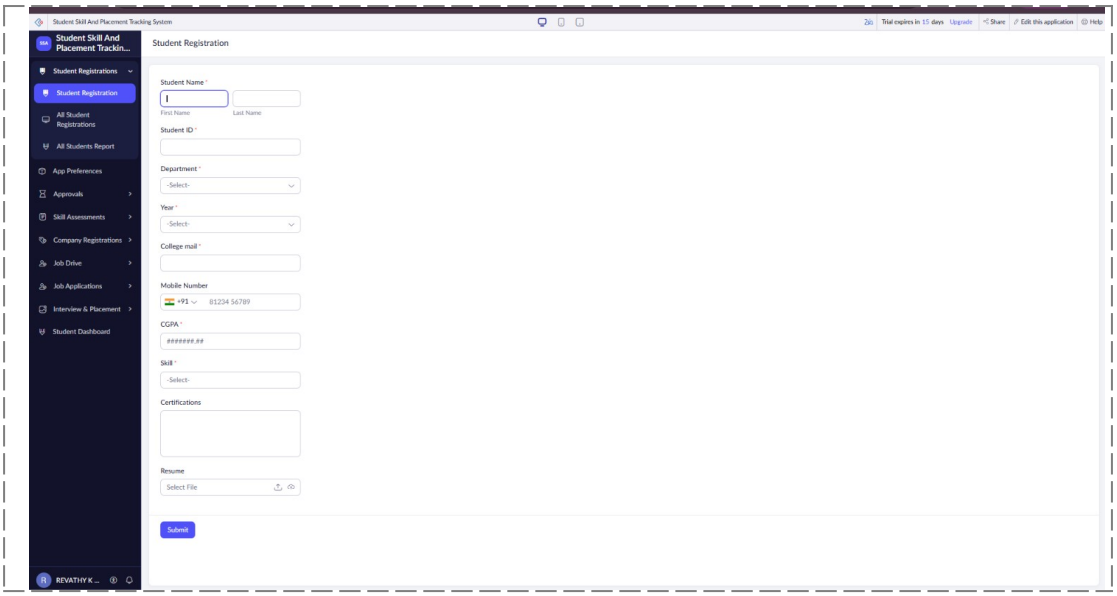


Figure 5.3: Student Registration Form in Live Mode

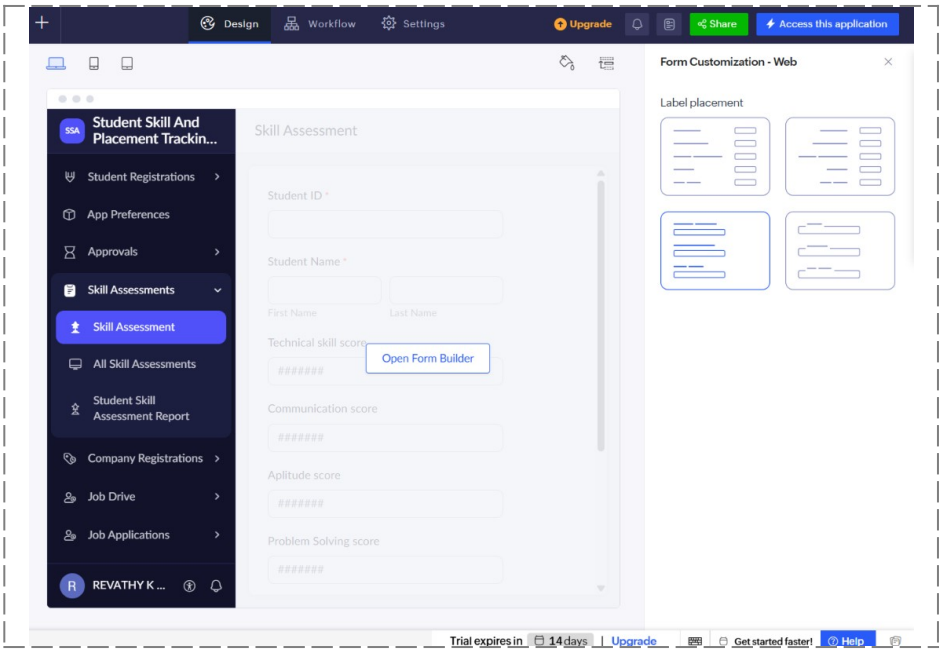


Figure 5.4: Skill Assessment Form in Design Mode (Form Customization – Web)

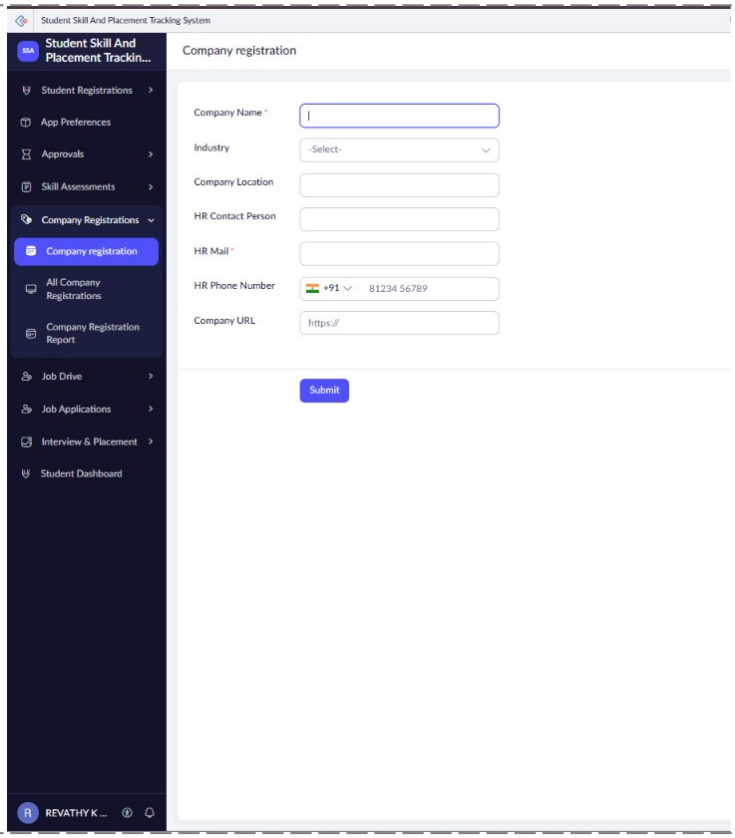


Figure 5.5: Company Registration Form in Live Mode

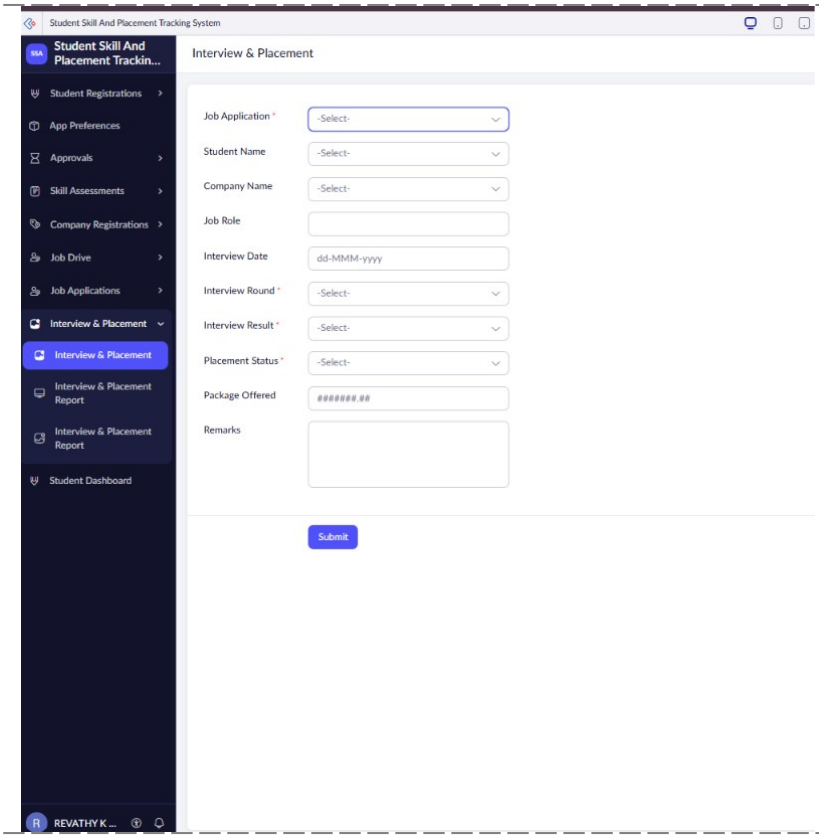


Figure 5.6: Interview & Placement Form in Live Mode

5.3 Reports

Seven reports present the data of the six forms: the registration, skill assessment and company records, the job drive and job application records, and the interview and placement results. They are listed in Table 5.1, and two of them are shown in Figure 5.7 and Figure 5.8. Reports are opened from the form sections of the left-hand menu, and the Student Dashboard has links to the main reports.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Student Registrations	List	Student Registration	Search and column sorting; no filter or grouping applied	None	Admin, Placement Officer, Faculty; Student (own)

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Company Registrations	List	Company registration	Search and column sorting; columns Company ID, Company Name, Industry, Company Location, HR Contact Person, HR Mail, HR Phone Number	None	Admin, Placement Officer, Faculty
All Skill Assessments	List	Skill Assessment	Search and column sorting; no filter or grouping applied	None	Admin, Placement Officer, Faculty; Student (own)
Student Skill Assessment Report	Card layout	Skill Assessment	Cards showing student and scores	None	Admin, Placement Officer, Faculty; Student (own)
Job Drive Report	List	Job Drive	Search and column sorting; no filter or grouping applied	None	Admin, Placement Officer, Faculty, Student
Job Applications Report	List	Job Application	Search and column sorting; columns Application ID, Student Name, Job Drive, Application Date, Application Status	None	Admin, Placement Officer, Faculty; Student (own)
Interview & Placement Report	List	Interview & Placement	Search and column sorting; no filter or grouping applied	None	Admin, Placement Officer, Faculty; Student (own)

Filters, grouping and custom actions were not configured on these reports; each report uses the default search and column sorting of Zoho Creator.

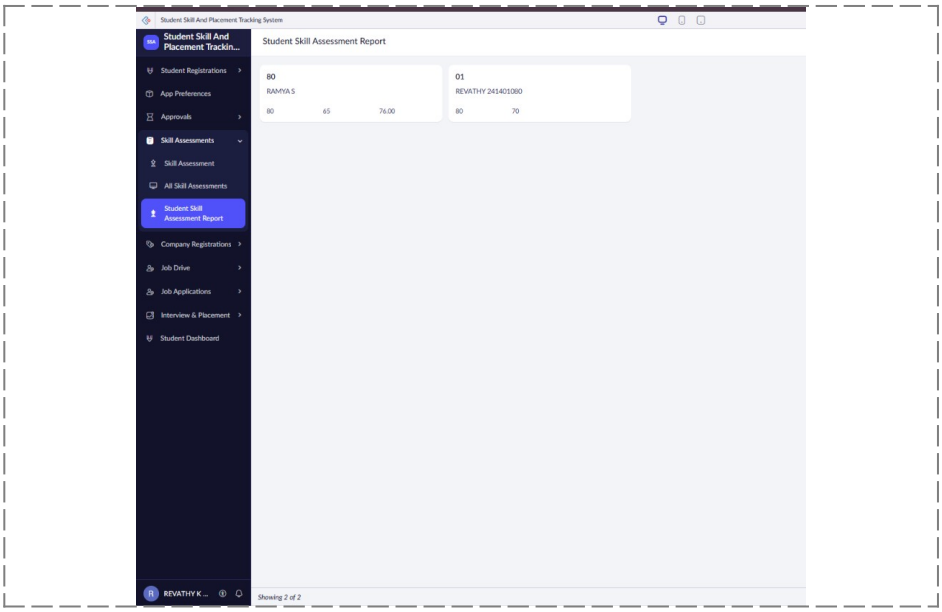


Figure 5.7: Student Skill Assessment Report (2 of 2 records)

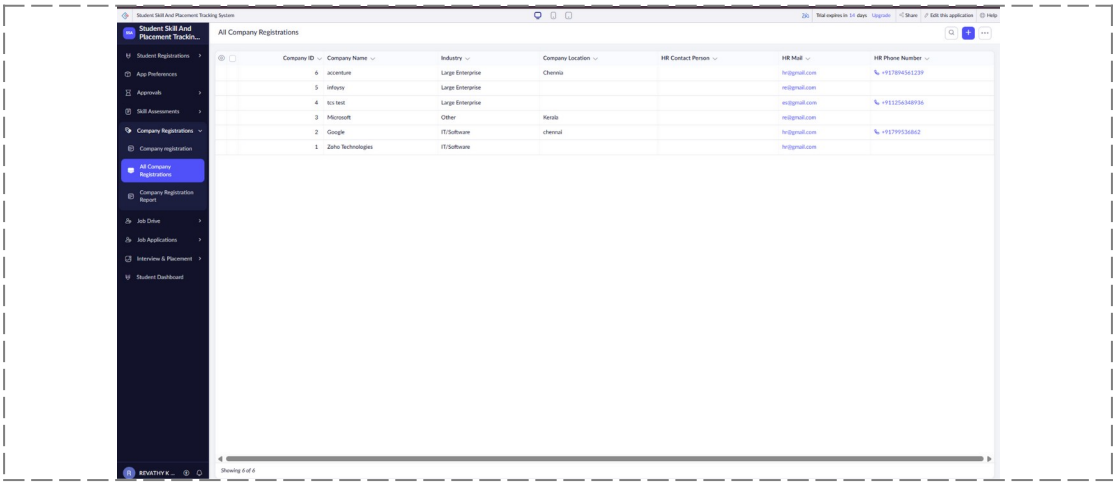


Figure 5.8: All Company Registrations Report (List, 6 of 6 records)

5.4 Pages and Dashboards

The Student Dashboard is the home page of the application and is built with the page builder for the Admin, Placement Officer and Faculty roles. It has a welcome banner with two shortcut buttons (View All Students and View Applications), a Quick Actions row with shortcuts to the Student Registration, Company Registration, Job Drive and Job Application forms, and five KPI panels: Registered Students, Registered Companies, Job Drives, Job Applications and Placed Students. Below them are three charts, Students by Department (bar chart), Companies by Industry (pie chart) and Placement Status (pie chart), and a list of links to the main reports. The page is shown in Figure 5.1.

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Company registration → Created or Edited → Successful form submission (workflow Sync Company to Zoho CRM)	Create the company as an Account in Zoho CRM	Map, put(), ifnull(), zoho.crm.createRecord() with a connection
DS-02	Job Application → Created → Successful form submission (workflow Job Eligibility Check)	Shortlist or reject an application by comparing CGPA with the Minimum CGPA	Fetch records by lookup ID, if / else, field assignment
DS-03	Interview & Placement → Edited → Successful form submission (workflow Placement Status Notification)	E-mail a placement confirmation to the student	if, fetch record, count(), sendmail

DS-01 (Sync Company to Zoho CRM) runs when a Company registration record is created or edited [3]. It builds a map with the Account_Name, Industry and Phone of the company and a Description made of its location, HR contact person and HR mail, and then calls zoho.crm.createRecord() on the Accounts module through the connection zoho_crm_connection (Figure 5.9). The complete code is in Appendix C.1.

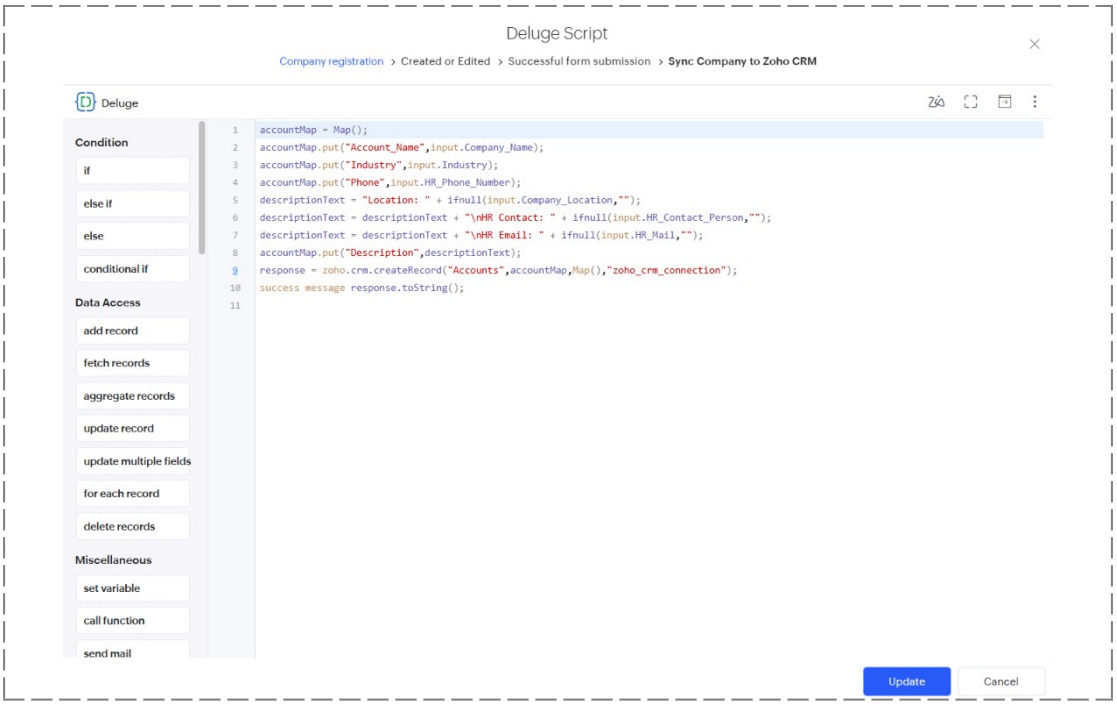


Figure 5.9: Deluge Editor – DS-01 Sync Company to Zoho CRM

DS-02 (Job Eligibility Check) runs when a Job Application is created. It fetches the selected Job Drive and the selected Student Registration record through their lookup IDs, compares the CGPA of the student with the Minimum_CGPA of the drive, and sets Application_Status to Shortlisted when the CGPA is equal or higher and to Rejected otherwise (Figure 5.10). The complete code is in Appendix C.2.

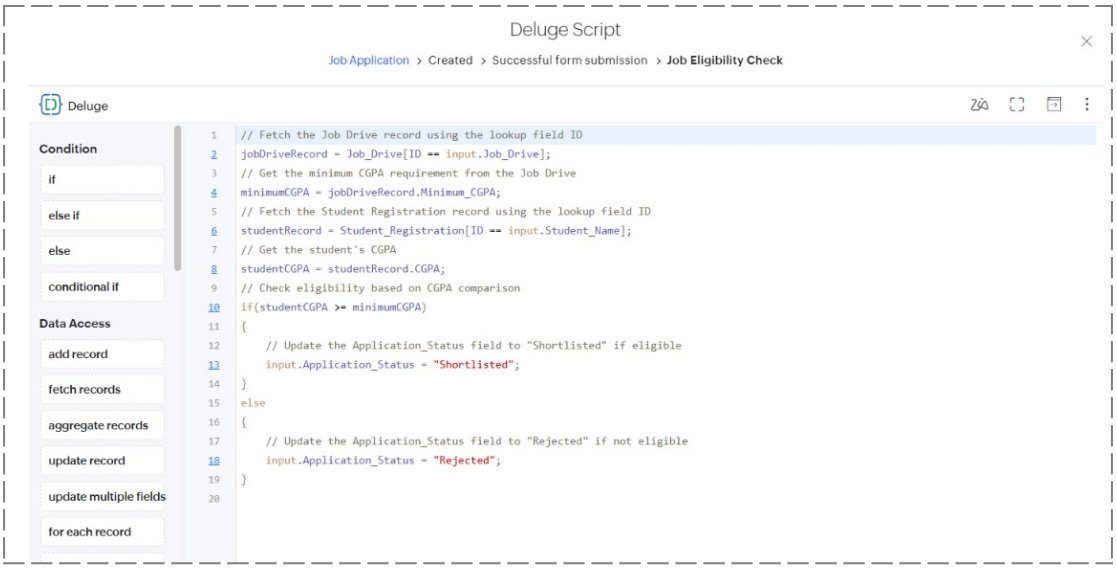


Figure 5.10: Deluge Editor – DS-02 Job Eligibility Check

DS-03 (Placement Status Notification) runs when an Interview & Placement record is edited. If Placement_Status is Placed, it fetches the record of the student and sends a Placement Confirmation e-mail to the college mail of the student with sendmail (Figure 5.11). The complete code is in Appendix C.3.

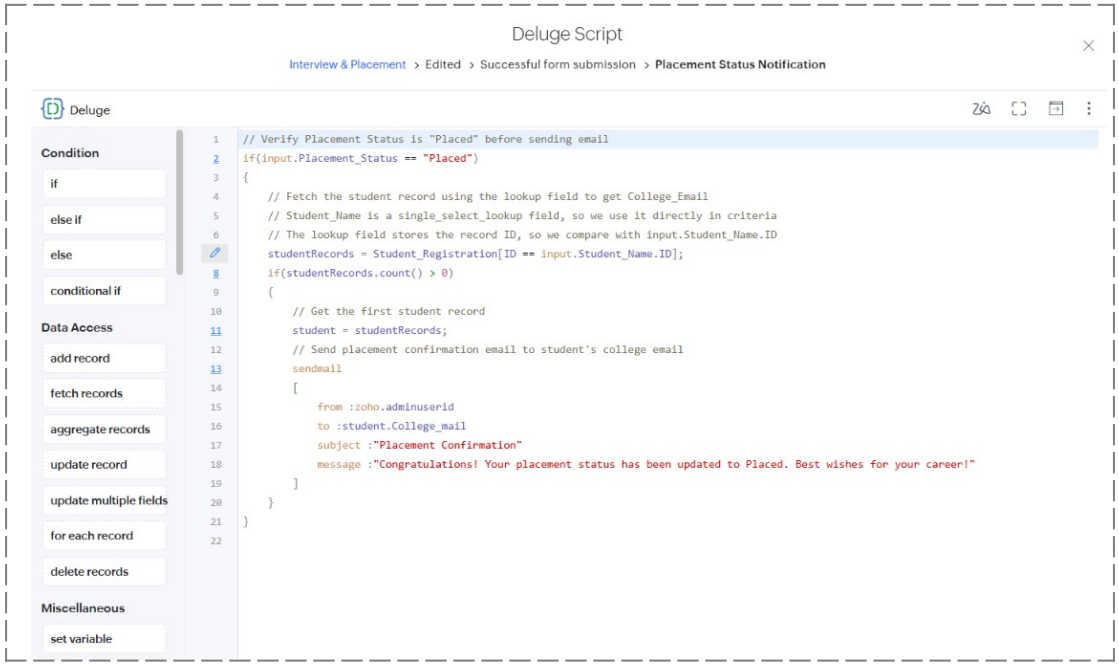


Figure 5.11: Deluge Editor – DS-03 Placement Status Notification

5.6 Workflows and Automation

The Workflows page lists the form workflows and the other workflow types. Four forms have workflows: Company registration (Sync Company to Zoho CRM, created or edited), Job Application (Job Eligibility Check, created), Interview & Placement (Placement Status Notification, edited) and Skill Assessment; Job Drive and Student Registration have none (Figure 5.12). One schedule workflow, Interview Reminder, runs on the Interview Date of the Interview & Placement form, has one action and is enabled (Figure 5.13). The workflows are listed in Table 5.3. The Skill Assessment Workflow is a form workflow that runs when a skill assessment is submitted and calculates the overall score from the four scores. The approval process is listed in the Approvals section of the menu and holds a submitted record until the approver acts on it.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Sync Company to Zoho CRM	Form Workflow	Company registration: created or edited, on successful form submission	—	Create the company as an Account in Zoho CRM	DS-01
Job Eligibility Check	Form Workflow	Job Application: created, on successful form submission	Student CGPA >= Minimum CGPA of the drive	Set Application_Status to Shortlisted, otherwise Rejected	DS-02
Placement Status Notification	Form Workflow	Interview & Placement: edited, on successful form submission	Placement_Status = Placed	Send the Placement Confirmation e-mail to the student	DS-03
Interview Reminder	Schedule	Interview & Placement: on the Interview Date	On the Interview Date, at the configured time	1 action (reminder)	No script (configured action)
Skill Assessment Workflow	Form Workflow	Skill Assessment: created, on successful form submission	—	Calculate the overall score from the four skill scores	—
Approval process (Approvals menu section)	Approval	Configured under the Approvals section of the menu (Figure 5.1)	—	Hold the record for approval by the approver	—

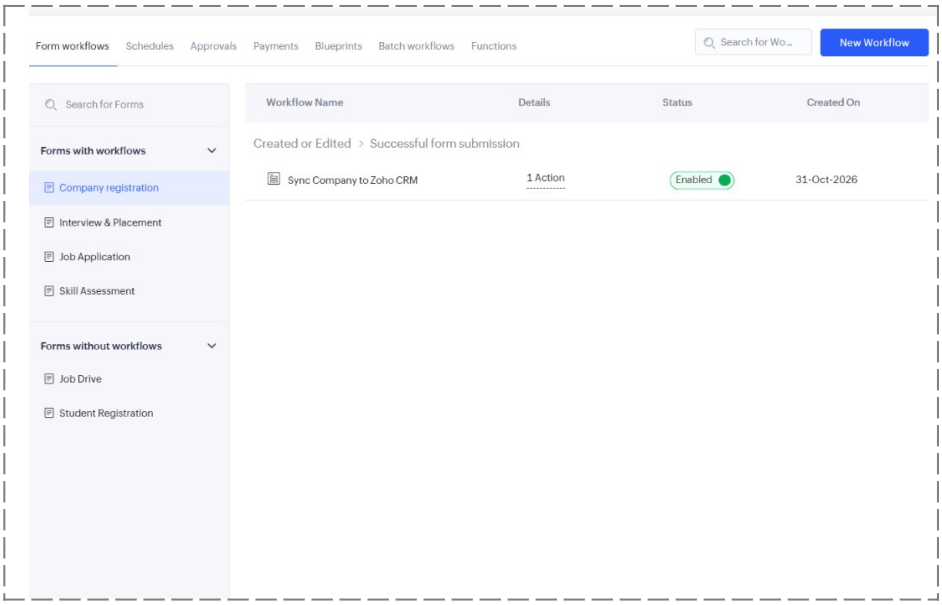


Figure 5.12: Form Workflows Page Showing Sync Company to Zoho CRM

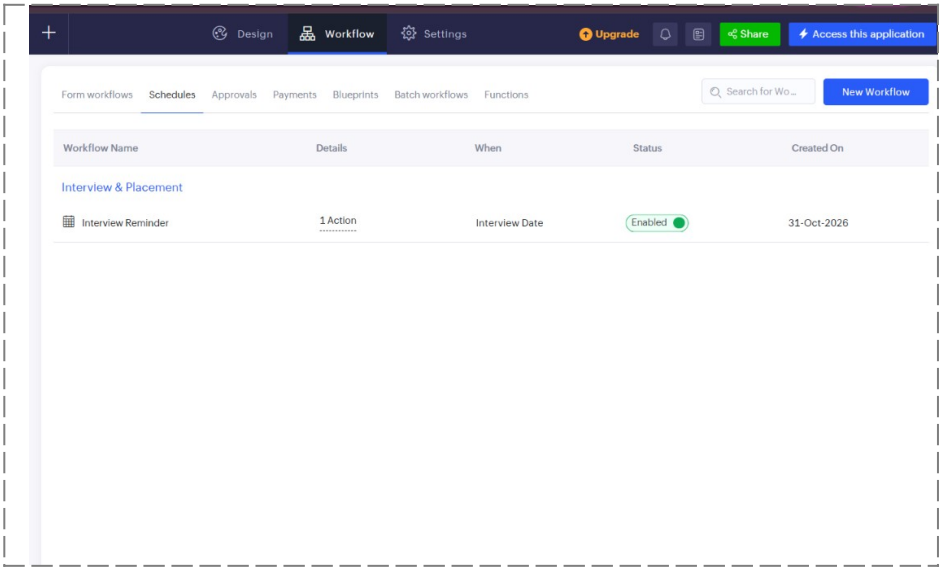


Figure 5.13: Schedule Workflow – Interview Reminder (Enabled)

5.7 Integrations

The application is integrated with Zoho CRM. When a company is registered, the workflow Sync Company to Zoho CRM creates the company as an Account in Zoho CRM through the connection zoho_crm_connection, which is shown as Connected on the Microservices → Connections page of Zoho Creator (Figure 5.14). Figure 5.15 shows the company accenture, registered in the application, as an Account in Zoho CRM. The integration is summarised in Table 5.4.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Company synchronisation	Zoho CRM (Accounts module)	Deluge zoho.crm.createRecord( ) with connection zoho_crm_connection (DS-01)	OAuth 2.0 (Zoho CRM connection)	Company name, industry, phone, location, HR contact and HR mail as a CRM Account

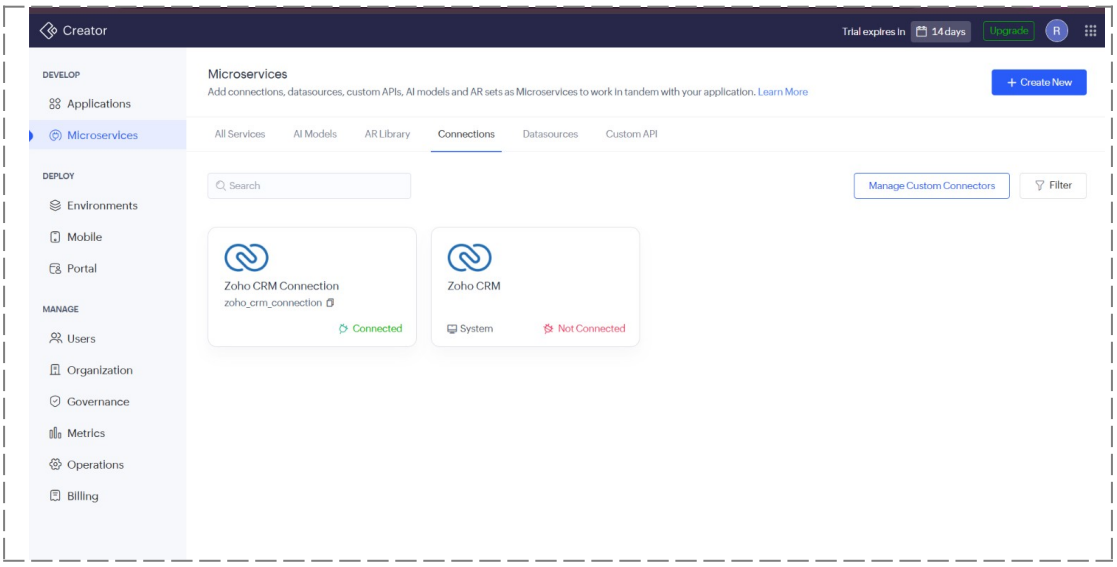


Figure 5.14: Zoho CRM Connection in Zoho Creator (Connected)

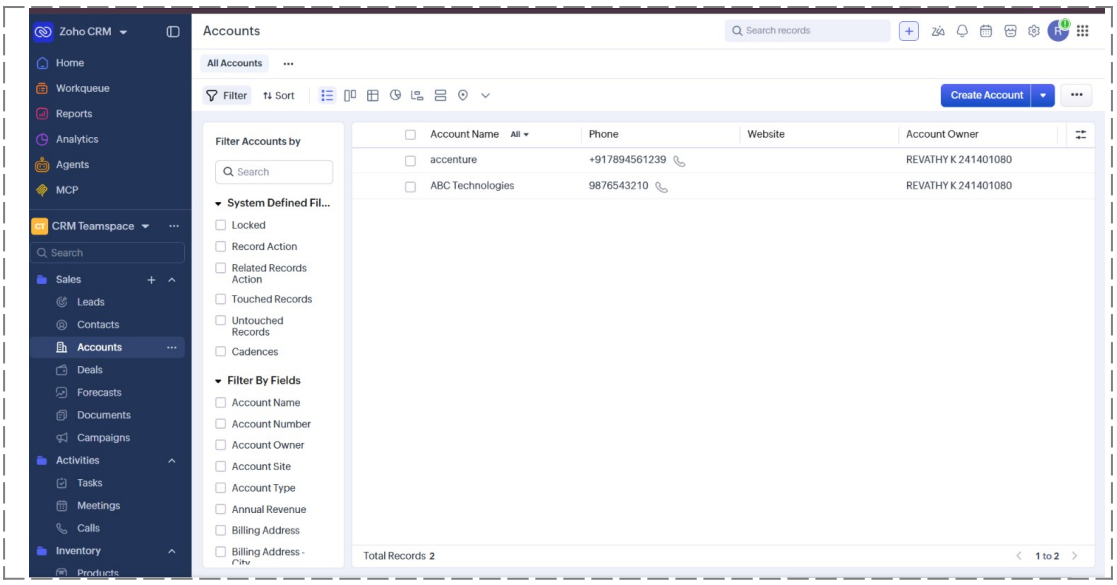


Figure 5.15: Synchronised Company (accenture) in Zoho CRM Accounts

5.8 User Management and Role-Based Access

Access to the application is controlled through roles and permission profiles. The role hierarchy has CEO at the top, with Admin, Faculty and Placement Officer below it, and Student reporting to Faculty (Figure 5.16). Four permission profiles were created – Admin Access, Faculty Access, Placement Officer Access and Student Access – in addition to the default Read and Write profiles. Each user is given a role and a profile, which decide the forms, reports and pages the user can open and the operations (add, edit, view, delete) allowed, as designed in Table 4.8.

No data sharing rules or record-level criteria were configured in this version. The permission profiles decide which forms, reports and pages each role can open, and the "own record only" view for students in Table 4.8 is a design target that is listed as a future enhancement. A customer portal is not used.

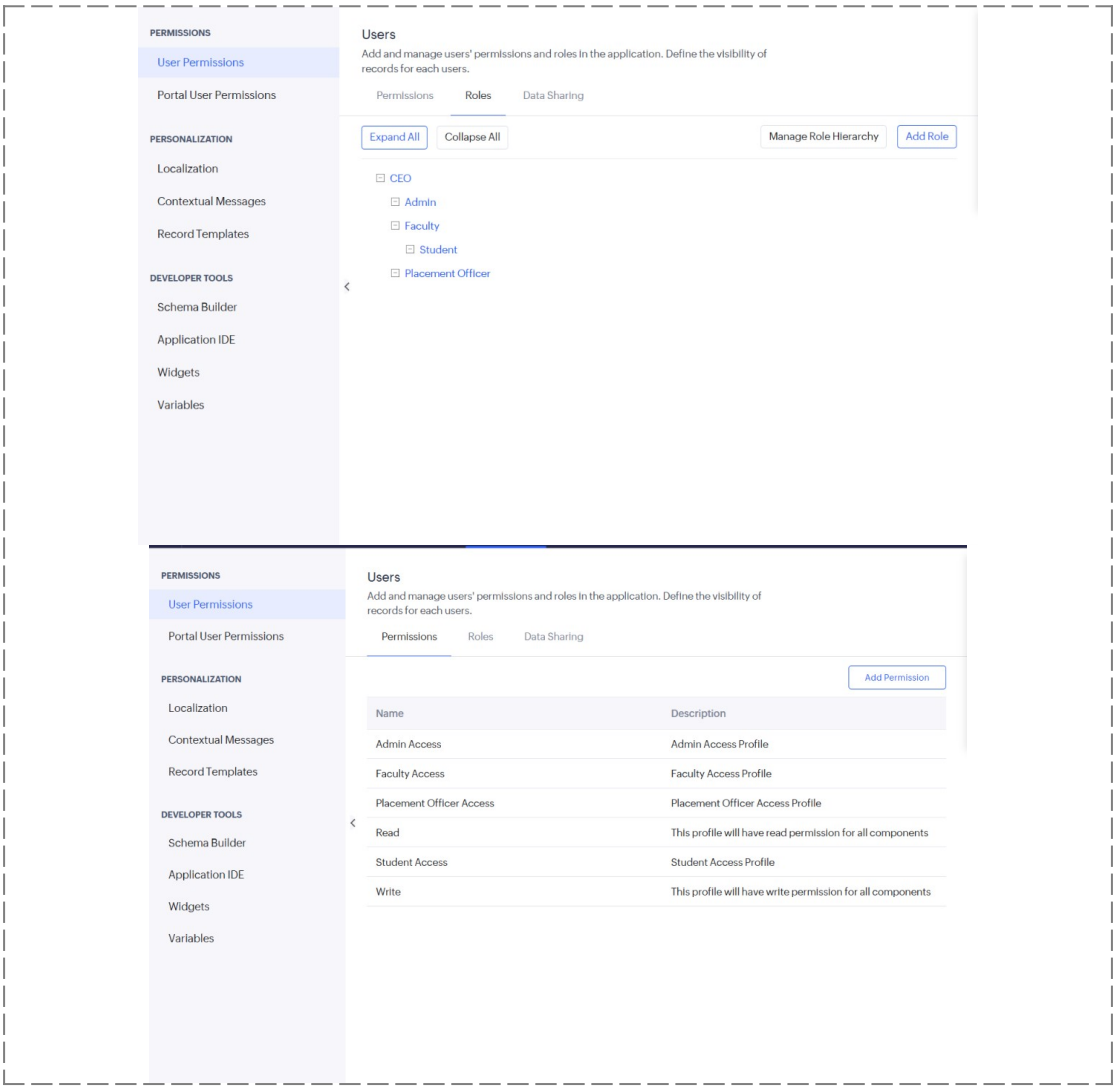


Figure 5.16: Role Hierarchy and Permission Profiles of the Application

5.9 AI and Analytics Features

Analytics: the Student Dashboard shows five KPI panels and three charts, Students by Department (bar chart), Companies by Industry (pie chart) and Placement Status (pie chart), which update as records are added (Figure 5.17).

No Zia or AI feature was used in this version; AI-based prediction is listed as a future enhancement in Section 9.2.

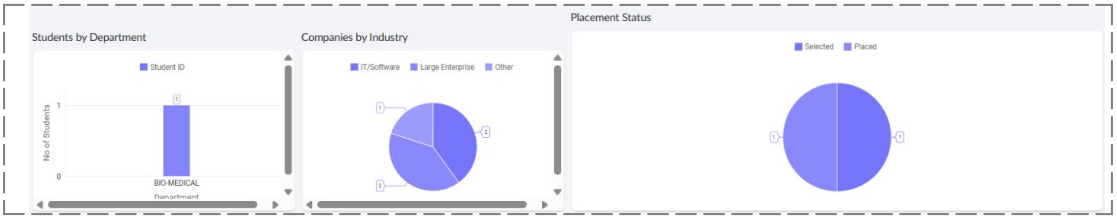


Figure 5.17: Analytics Charts on the Student Dashboard

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst function, widget or custom JavaScript) was used in this version. All requirements were met with forms, reports, workflows and short Deluge scripts. A pro-code extension would be considered for resume parsing, bulk import of student data or predictive placement scoring, which need more computation than Deluge handles comfortably.

5.11 Mobile and Portal Access

The application can be opened in the Zoho Creator mobile app with the same roles and permissions. The application was checked in the phone view of Zoho Creator (Figure 5.18): the Student Dashboard, the All Student Registrations report and the Company registration form adapt to the narrow screen. The dedicated Zoho Creator mobile app and a customer portal were not set up.

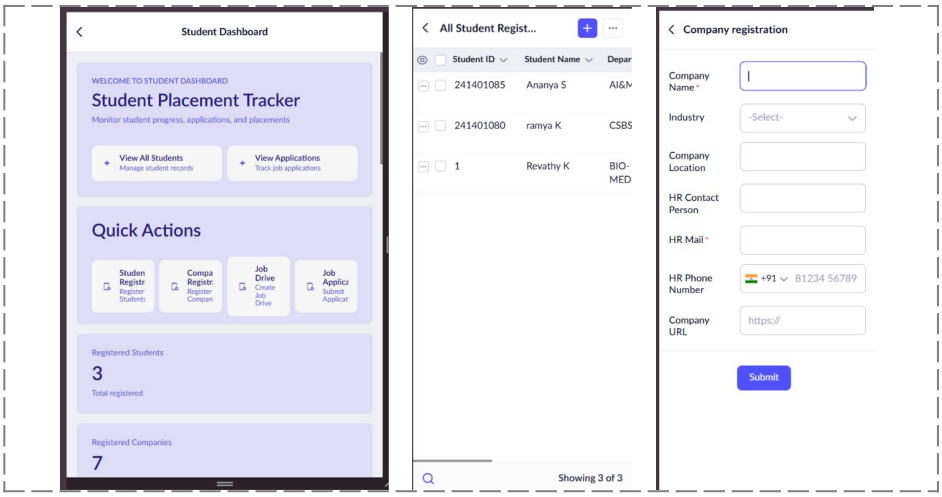


Figure 5.18: Application in Phone View: Student Dashboard, All Student Registrations and Company Registration

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. Unit testing checks each form: mandatory fields, field types and validation messages. Integration testing checks the workflows, the Zoho CRM synchronisation and the flow of data from application to approval to interview and placement. Role-based access testing is done by signing in as each role (Admin, Placement Officer, Faculty and Student) and verifying that only the permitted forms, reports and pages are available. User acceptance testing is done by asking end users to use the application in their role. Testing was carried out by the developer, REVATHY K, on the live application in Zoho Creator up to 01 November 2026, the date shown on the application records and screenshots in this report.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit Student Registration with all mandatory fields	Fill all mandatory fields and submit	Record created; success message	Record created; success message shown	Pass
TC-02	FR-01	Submit Student Registration without college mail	Leave college mail empty	Mandatory-field error; record not saved	Mandatory-field error shown; record not saved	Pass
TC-03	FR-01	Enter an invalid email address	College mail = “abc”	Invalid email message	Invalid email message shown	Pass
TC-04	FR-01	Submit Student Registration without CGPA	Leave CGPA empty	Mandatory-field error	Mandatory-field error shown	Pass
TC-05	FR-02	Record a skill assessment for a registered student	Enter student ID, name and the four scores, submit	Record created; shown in Student Skill Assessment Report	Record created; shown in Student Skill Assessment Report	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-06	FR-03	Register a company with mandatory fields	Company name and HR mail entered	Record created; Company ID generated	Record created; Company ID generated	Pass
TC-07	FR-03	Register a company without HR mail	Leave HR mail empty	Mandatory-field error	Mandatory-field error shown	Pass
TC-08	FR-04	Company synchronisation to Zoho CRM	Submit a new company	Account appears in Zoho CRM Accounts	Account created in Zoho CRM (accenture, Figure 5.15)	Pass
TC-09	FR-05	Create a job drive with a Minimum CGPA	Select company, enter Minimum CGPA	Drive saved and listed in Job Drive Report	Drive saved and listed in Job Drive Report	Pass
TC-10	FR-06	Submit a job application	Select student and job drive	Application created	Application created (Application ID 1, Revathy K, java developer)	Pass
TC-11	FR-07	Eligible student applies	Student CGPA >= Minimum CGPA	Application_Status = Shortlisted	Application_Status set to Shortlisted (Figure 8.2)	Pass
TC-12	FR-07	Non-eligible student applies	Student CGPA < Minimum CGPA	Application_Status = Rejected	Application_Status set to Rejected	Pass
TC-13	FR-08	Record interview and placement	Select job application, enter round, result, status, package	Record saved	Record saved and listed in Interview & Placement Report	Pass
TC-14	FR-08	Submit Interview & Placement without interview result	Leave Interview Result empty	Mandatory-field error	Mandatory-field error shown	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-15	FR-09	Placement confirmation e-mail	Edit a record and set Placement Status = Placed	Student receives the “Placement Confirmation” e-mail	Placement Confirmation e-mail sent to the student	Pass
TC-16	FR-10	Dashboard refresh after adding records	Add a student and a company	KPI panels and charts show the new counts	KPI panels and charts show the new counts	Pass
TC-17	FR-11	Interview Reminder schedule runs	Set an Interview Date and wait for the schedule	Reminder sent on the Interview Date as configured	Reminder sent on the Interview Date	Pass
TC-18	FR-12	Student opens the Company Registration report	Sign in as Student	Access denied or report not visible	Company Registration report not available to the Student role	Pass
TC-19	FR-12	Faculty tries to delete a record	Sign in as Faculty and try to delete	Operation not permitted	Delete operation not available to the Faculty role	Pass

6.3 Debugging Deluge Scripts

Deluge scripts were debugged with info statements that print variable values, the execute option of the Deluge editor, and the error messages shown on form submission; workflow runs were checked in the workflow logs. One real bug was met in DS-03: the placement e-mail was not sent because the student record was fetched by comparing the Student_Name lookup with the name instead of its record ID, and the e-mail field was referred to by a wrong link name (College_Email instead of College_mail). The criteria were changed to ID == input.Student_Name.ID and the field College_mail was used, as in Appendix C.3.

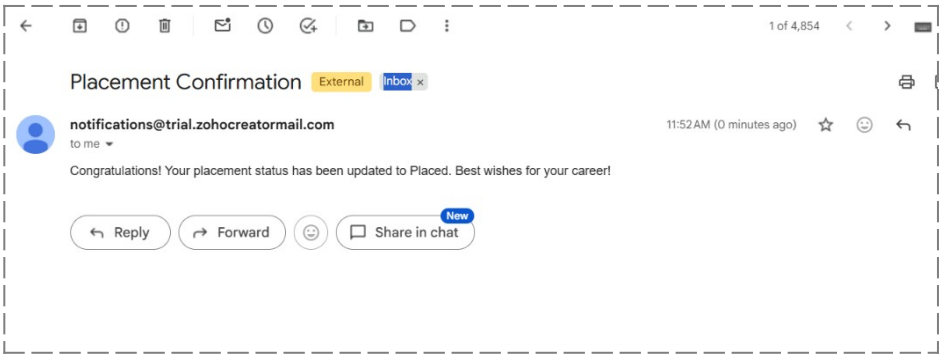


Figure 6.1: Placement Confirmation E-mail Received after DS-03 Ran Successfully

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Placement Confirmation e-mail not sent when a record was set to Placed	Student record fetched by comparing the lookup with the student name instead of the record ID; wrong link name used for the e-mail field	Criteria changed to ID == input.Student_Name.ID and field changed to College_mail in DS-03	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
REVATHY K (developer)	Admin (all forms, reports and the dashboard)	Forms are easy to fill and the eligibility result appears as soon as an application is submitted	None required
No other users	—	No feedback from other users was collected; testing with the Placement Officer and Faculty is planned after the demonstration	Listed in Section 9.2

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. Changes are made in Development, checked in Stage and then published to Production, and version notes allow a rollback if needed. The application was built and tested in the Development environment of the trial edition, checked in Stage and then published to Production as version 1.0 on 01 November 2026. Figure 7.1 shows version 1.0 in both the Stage and Production columns of the Environments page.

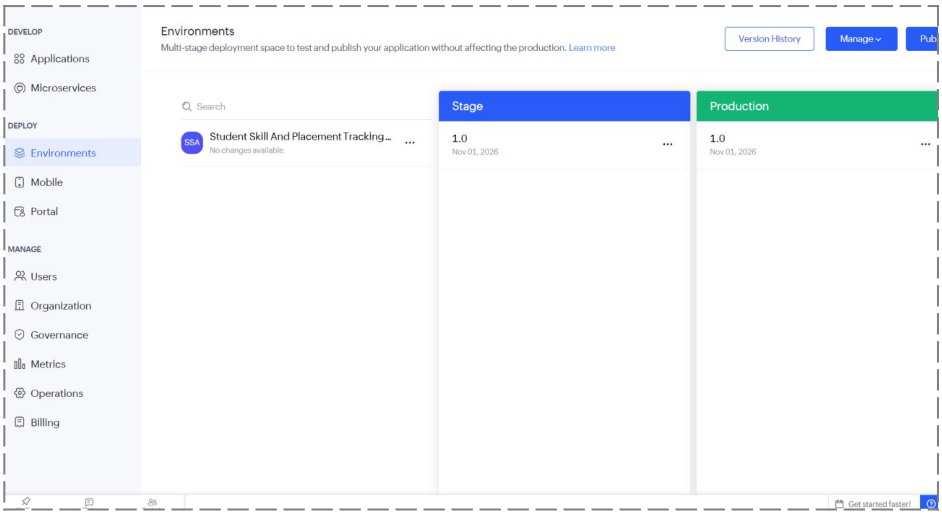


Figure 7.1: Environments Page and Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Performance was kept in mind while building the application: reports show only the columns that are needed, a fetch with criteria is preferred to looping over all records, and filters limit the number of records displayed. No timing measurements were taken, so no improvement is claimed; Table 7.2 records the design choices that help performance.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Reports show only the needed columns	All reports	—	—	Not measured
Fetch by lookup ID instead of looping over records	DS-02, DS-03	—	—	Not applicable (load time not measured)

7.3 Security Features

The following security controls are used in the application and are listed in Table 7.3.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permission profiles (Admin, Faculty, Placement Officer, Student Access)	User Permissions → Permissions and Roles	Unauthorised access to forms, reports and pages
OAuth 2.0 connection to Zoho CRM	Workflow: Sync Company to Zoho CRM	Credential leakage (no keys written in scripts)
Mandatory fields and field-type validation	Student Registration, Company registration, Interview & Placement	Incomplete or invalid data
Personal-data (PII) flag on student fields	Student Registration form (e.g. Student Name)	Exposure of personal data
Record-level criteria / Data Sharing rules (not applied)	Not configured in this version (Section 9.2)	Unauthorised access to other users' records

7.4 Governance and Administration

Governance relies on the built-in features of Zoho Creator: the audit trail and record history, controlled user and role management by the Admin, consistent naming of forms, reports and workflows, and the ability to export data for backup. Administration is done in the application settings (users, roles and permission profiles) by the Admin, who owns and maintains the application. Forms are named after the entity, reports after the form or purpose (for example Job Application Report) and workflows after what they do.

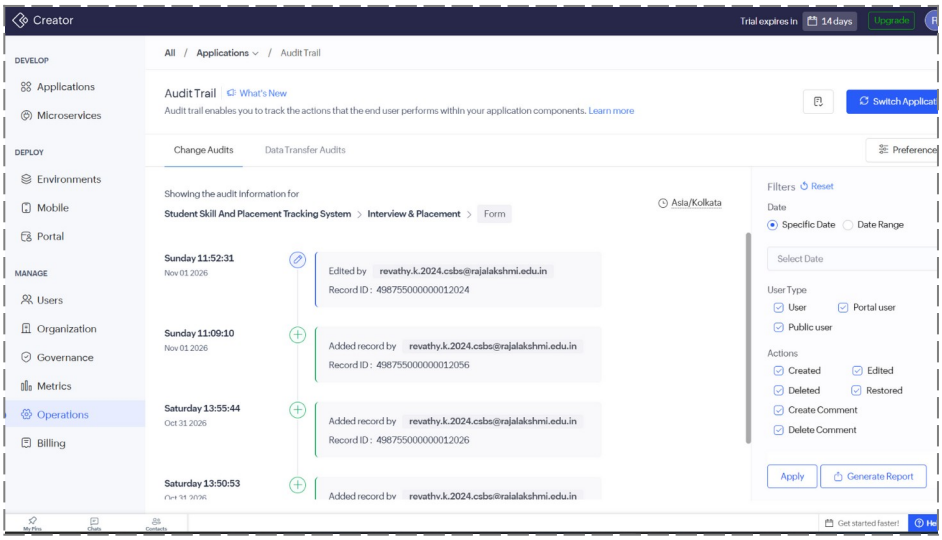


Figure 7.2: Audit Trail and Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario shows the application end to end. The Placement Officer registers a company, and the workflow Sync Company to Zoho CRM creates it as an Account in Zoho CRM. The Placement Officer creates a job drive for the company. A student registers and submits a job application for the drive, and the eligibility check sets its status to Shortlisted or Rejected. After the interview, the Placement Officer records the round, result, placement status and package in the Interview & Placement form, a placement confirmation e-mail is sent when the status is Placed, and the Student Dashboard and reports show the updated figures (Figure 5.1). Figure 8.2 shows step 2: the application of Revathy K for the java developer drive, submitted on 01-Nov-2026 and set to Shortlisted by the eligibility check. Figure 8.4 shows the Placement Confirmation e-mail that the student received after the placement status was set to Placed.

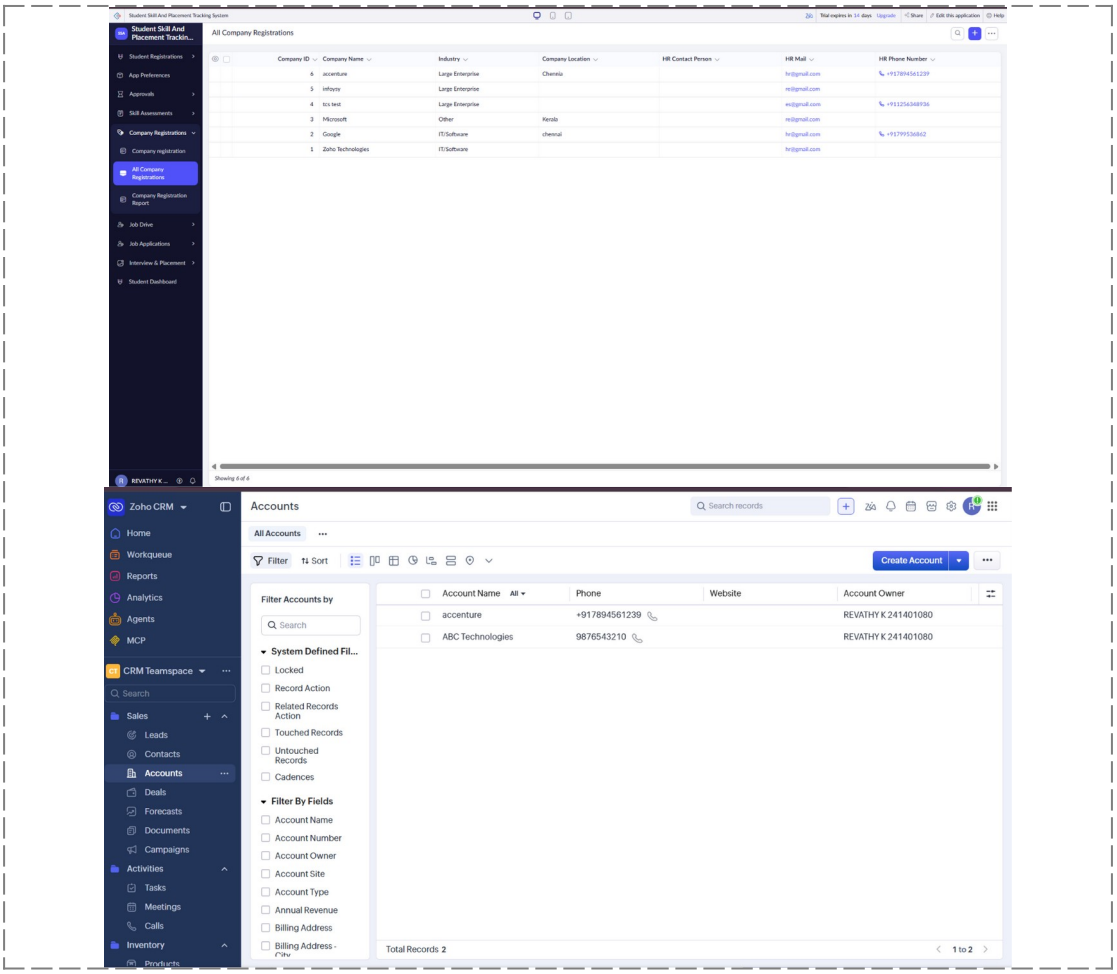


Figure 8.1: Scenario Step 1 – Company registered and synchronised to Zoho CRM

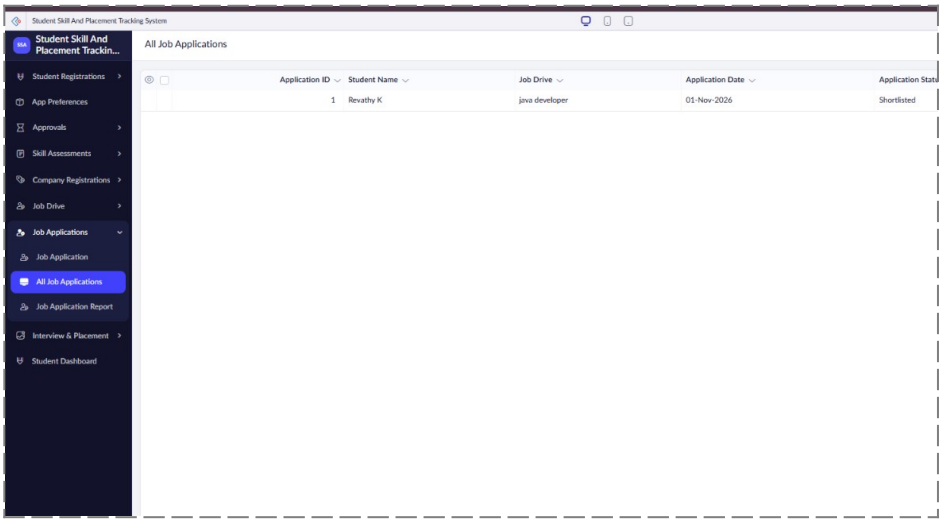


Figure 8.2: Scenario Step 2 – Student application submitted and eligibility checked

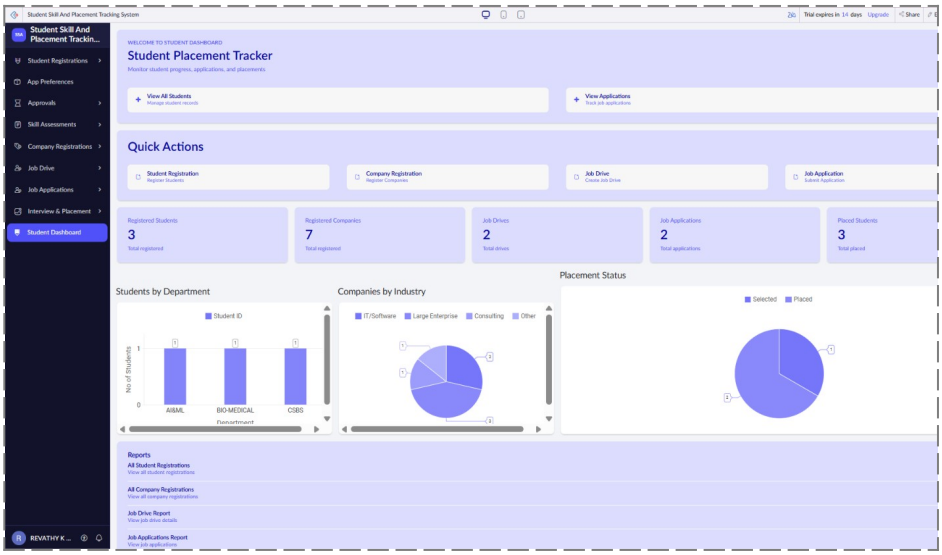


Figure 8.3: Scenario Step 3 – Interview and placement recorded; dashboard updated

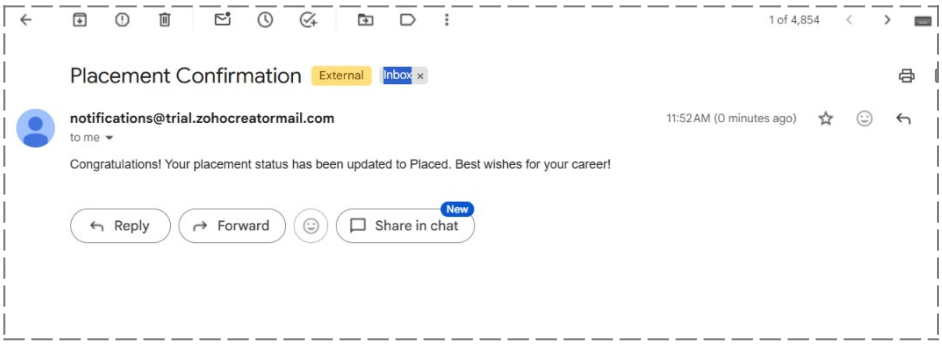


Figure 8.4: Placement Confirmation E-mail Received by the Student

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 1 – Data model and forms	Sections 4.2, 4.3 and 5.2	Six connected forms (Figure 5.3, Figure 5.5, Figure 5.6)
Objective 2 – Automation with workflows and Deluge	Sections 5.5 and 5.6	Company synchronisation (Figure 5.9), job eligibility check (Figure 5.10), placement e-mail (Figure 5.11) and interview reminder (Figure 5.13)
Objective 3 – Role-based dashboards and reports	Sections 5.3, 5.4 and 5.8	Student Dashboard (Figure 5.1), seven reports, roles and profiles (Figure 5.16)
Objective 4 – Integration with Zoho CRM	Section 5.7	Connection (Figure 5.14) and synchronised company (Figure 5.15)

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 5 – Secure deployment and governance	Sections 7.1 to 7.4	Sections 7.1 to 7.4 and Table 7.3 (permission profiles and OAuth connection); Figure 7.1 (version 1.0 in Stage and Production)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Entry in spreadsheets and emails, followed by merging	One online form; the record is available at once	Faster, with no re-entry
Manual steps	Re-typing, merging sheets and checking eligibility by hand	Lookups, workflows and approval process	Fewer manual steps
Report generation	Prepared by hand	Live reports and dashboard	Real-time information
Data accessibility	Files shared by email, no access control	Role-based access from any browser or mobile app	Controlled access from anywhere

These comparisons are qualitative because the two processes were not timed.

8.4 Limitations of the Low-Code Approach

Zoho Creator suits form-and-workflow applications such as this one, but some limits were kept in mind, in line with studies of low-code practice [6], [7]. Customisation of pages and dashboards is limited to the components the platform provides, so exact visual layouts are harder than in a hand-coded web page. Deluge has limits on statements and API calls in one execution, which matters for bulk operations such as importing a whole batch of students. Complex analytics and prediction are better done in a dedicated analytics or machine-learning service. The application also depends on the Zoho platform and its plan limits, which is a form of vendor lock-in.

Editions and trial limits: the application was built on a trial edition (14 days remaining on 01 November 2026), so some governance features could not be assumed. Lookup display: a lookup shows only the field chosen as its display value, so a drive appears as

its job role (for example java developer) without the company name; a concatenated display field would give a more informative label.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the no-code → low-code → pro-code spectrum, most of the application is no-code configuration: forms, reports, the Student Dashboard page, roles and permission profiles, and the schedule workflow. Low-code is used where the logic needs a script: the company synchronisation to Zoho CRM, the job eligibility check and the placement notification e-mail. No pro-code component was needed. A move to pro-code, for example with Zoho Catalyst, would be justified for resume parsing, predictive scoring of placement readiness or large bulk imports.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project built a complete placement tracking application on Zoho Creator for the Department of Computer Science and Business Systems. Six connected forms capture students, skill assessments, companies, job drives, job applications and interview and placement results; seven reports and the Student Dashboard give live information; form workflows synchronise companies to Zoho CRM, check job eligibility by CGPA and e-mail placement confirmations; a schedule workflow sends interview reminders; and four roles with permission profiles control access.

The work shows that a business process that was managed in scattered spreadsheets can be moved to a single connected application quickly using low-code tools, with the platform providing hosting, security and mobile access. Practitioner studies report challenges such as customisation and integration limits [6], [7], and the limits of the approach, such as page-builder customisation and execution limits, were within acceptable bounds for this application.

9.2 Future Enhancements

- Predictive placement-readiness scoring of students using Zia or a Catalyst-based model.
- Email and SMS notifications to students for new drives, shortlists and results.
- Automatic resume parsing and skill-gap analysis against the skills required by each drive.
- A customer portal for companies to view applicants and update interview results.
- A Catalyst-based microservice for bulk import of student data and heavy reporting.

9.3 Learning Outcomes and Career Relevance

Developing this application gave practical experience in data modelling with forms and lookups, workflow and approval design, Deluge scripting, integration with Zoho CRM, and role-based access control. These skills relate directly to roles such as citizen developer, low-code developer, solutions consultant and automation engineer, and the

understanding of where low-code ends and pro-code begins helps in choosing the right tool for a business problem.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Student Skill And Placement Tracking System – menu sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Student Registration, Company registration, Interview & Placement forms	5.2	Figure 5.3, Figure 5.5, Figure 5.6
II	Lookup field (relationship between forms)	Y	Job Application (Job Drive, Student Name); Interview & Placement (Student Name)	4.2, 5.2	Figure 5.6
II	Subform (parent–child data)	N	Not used; lookup fields connect the forms	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields on Student Registration, Company registration, Interview & Placement	5.2	Figure 5.2
II	List report with filter, grouping, custom action	N	List reports exist but no filter, grouping or custom action was configured	5.3	Figure 5.8
II	Kanban / Calendar / Timeline / Map report	N	Not used (list and card reports only)	5.3	—
II	Pivot table / chart report	Y	Student Dashboard charts (Students by Department, Companies by Industry, Placement Status)	5.3	Figure 5.17

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Page / dashboard built with UI Builder	Y	Student Dashboard page	5.4	Figure 5.1
II	Deluge – form events (validate / success / field actions)	Y	Form workflows on Company registration, Job Application, Interview & Placement	5.5	Figure 5.9, Figure 5.10, Figure 5.11
II	Deluge – functions, fetch, loops, collections	Y	Fetch records and if / else in DS-02 and DS-03	5.5	Figure 5.10
II	Email / in-app notifications	Y	sendmail in DS-03 (Placement Confirmation e-mail)	5.5, 5.6	Figure 5.11
II	Form workflow	Y	Sync Company to Zoho CRM (Company registration)	5.6	Figure 5.12
II	Scheduled workflow	Y	Interview Reminder (Interview & Placement)	5.6	Figure 5.13
II	Approval process	Y	Approval process listed in the Approvals menu section	5.6	Figure 5.1
II	Blueprint	N	Not used	5.6	—
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	Y	Test cases (Table 6.1) and defect log (Table 6.2)	6	Table 6.1
III	Integration with a Zoho service	Y	Zoho CRM – Sync Company to Zoho CRM	5.7	Figure 5.15
III	Connection / REST API call (invokeurl)	Y	Connection zoho_crm_connection used by DS-01	5.7	Figure 5.14
III	Webhook or Creator REST API	N	Not used	5.7	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Users, roles and permission profiles	Y	Roles Admin, Faculty, Student, Placement Officer; four permission profiles	5.8	Figure 5.16
III	Record-level access criteria / customer portal	N	Not configured (Section 5.8)	5.8	—
III	Zia / AI feature	N	No Zia or AI feature used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Student Dashboard charts and KPI panels	5.9	Figure 5.17
IV	Performance optimisation applied & measured	N	No measurements taken; design choices in Table 7.2	7.2	—
IV	Deployment across environments	Y	Version 1.0 published to Stage and Production on 01 Nov 2026	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Permission profiles; OAuth connection to Zoho CRM	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Admin settings for users, roles and permission profiles; naming conventions	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	No pro-code extension used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Phone view checked (Figure 5.18); mobile app and portal not set up	5.11	Figure 5.18

Total features implemented: 22 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Good	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Satisfactory	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Good	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Company registration – Sync Company to Zoho CRM (DS-01)

```
// Script ID : DS-01
// Location  : Company registration → Created or Edited → Successful
form submission (workflow: Sync Company to Zoho CRM)
// Purpose   : Create the company as an Account in Zoho CRM

accountMap = Map();
accountMap.put("Account_Name",input.Company_Name);
accountMap.put("Industry",input.Industry);
accountMap.put("Phone",input.HR_Phone_Number);
descriptionText = "Location: " + ifnull(input.Company_Location,"");
descriptionText = descriptionText + "\nHR Contact: " +
ifnull(input.HR_Contact_Person,"");
descriptionText = descriptionText + "\nHR Email: " +
ifnull(input.HR_Mail,"");
accountMap.put("Description",descriptionText);
response =
zoho.crm.createRecord("Accounts",accountMap,Map(),"zoho_crm_connection
");
success message response.toString();
```

C.2 Job Application – Job Eligibility Check (DS-02)

```
// Script ID : DS-02
// Location  : Job Application → Created → Successful form submission
(workflow: Job Eligibility Check)
// Purpose   : Shortlist or reject the application by comparing CGPA
with the Minimum CGPA

// Fetch the Job Drive record using the lookup field ID
jobDriveRecord = Job_Drive[ID == input.Job_Drive];
// Get the minimum CGPA requirement from the Job Drive
minimumCGPA = jobDriveRecord.Minimum_CGPA;
// Fetch the Student Registration record using the lookup field ID
studentRecord = Student_Registration[ID == input.Student_Name];
// Get the student's CGPA
studentCGPA = studentRecord.CGPA;
// Check eligibility based on CGPA comparison
if(studentCGPA >= minimumCGPA)
{
    // Update the Application_Status field to "Shortlisted" if
eligible
    input.Application_Status = "Shortlisted";
}
else
{
    // Update the Application_Status field to "Rejected" if not
eligible
    input.Application_Status = "Rejected";
}
```

C.3 Interview & Placement – Placement Status Notification (DS-03)

```
// Script ID : DS-03
// Location  : Interview & Placement → Edited → Successful form
submission (workflow: Placement Status Notification)
// Purpose   : E-mail a placement confirmation to the student when the
status is Placed

// Verify Placement Status is "Placed" before sending email
if(input.Placement_Status == "Placed")
{
    // Fetch the student record using the lookup field to get
College_Email
    // Student_Name is a single_select_lookup field, so we use it
directly in criteria
    // The lookup field stores the record ID, so we compare with
input.Student_Name.ID
    studentRecords = Student_Registration[ID ==
input.Student_Name.ID];
    if(studentRecords.count() > 0)
    {
        // Get the first student record
        student = studentRecords;
        // Send placement confirmation email to student's college
email
        sendmail
        [
            from :zoho.adminuserid
            to :student.College_mail
            subject :"Placement Confirmation"
            message :"Congratulations! Your placement status has been
updated to Placed. Best wishes for your career!"
        ]
    }
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link https://revathyk2024csbsrajal454.zohocreatorportal.in in a web browser, or open the application in the Zoho Creator mobile app. Sign in with the Zoho account that the Admin has assigned to your role. After sign-in the Student Dashboard opens; use the menu on the left to move between sections.

D.2 Instructions for Student

- Sign in and open Student Registrations → Student Registration.
- Enter your name, student ID, department, year, college mail, mobile number, CGPA and skill; add certifications and upload your resume; click Submit.
- Open Skill Assessments to view your skill assessment report.
- Open Job Applications → Job Application, select the job drive and click Submit.
- Open the Job Applications report to see whether each application is Shortlisted or Rejected.
- Open Student Dashboard to see the overall placement summary.

D.3 Instructions for Placement Officer

- Open Company Registrations → Company registration, enter the company details and click Submit; the company is synchronised to Zoho CRM automatically.
- Open Job Drive and create a drive for the company.
- Open the Job Applications report to see the status (Shortlisted or Rejected) that the eligibility check set for each application.
- Open Interview & Placement → Interview & Placement and record the interview date, round, result, placement status and package offered; an interview reminder is sent on the interview date, and a confirmation e-mail goes to the student when the status is Placed.
- Use the reports and the Student Dashboard to monitor students, companies, drives, applications and placements.

D.4 Instructions for Faculty

- Open Skill Assessments → Skill Assessment, select the student, enter the scores and click Submit.
- Open Student Skill Assessment Report and All Skill Assessments to review the skill development of students.
- Open Student Registrations → All Students Report to view student profiles.

D.5 Instructions for Admin

- Open the application settings → Users to add users and assign each user a role and a permission profile.
- Review all forms and reports, correct records where needed and check the audit trail.
- Check that the three form workflows and the Interview Reminder schedule are enabled.